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Techniques for adaptive frame-based clip matching

Abstract

A data processing system for matching video content implements obtaining first video footage that includes a plurality of first video frames of a first video clip; obtaining second video footage that includes a plurality of second video frames; analyzing the first video footage and the second video footage to generate a plurality of first fingerprints representing each frame of the plurality of first video frames and a plurality of second fingerprints representing the second video frames; comparing the first fingerprints to the second fingerprints to identify one or more candidate clip matches for the first video clip in the second video footage; selecting a best match for the first video clip in the second video footage from the one or more candidate clip matches for the first video clip; and presenting the best match for the first video clip on a user interface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority from U.S. Provisional Patent Application Ser. No. 63/081,318, filed on Sep. 21, 2020, and entitled “Adaptive Frame-based Clip Match.”

BACKGROUND

(1) Television and movie production typically generate large quantities of digital footage. Cameras may capture footage as high-definition video content, such a 4K or 8K resolution. Furthermore, multiple cameras may be used to capture the footage, which further increases the amount of high-resolution content that must be stored and processed for a particular streaming, television, or movie production. To process this content, the high-resolution content is typically transcoded to create a lower-resolution version of the footage to be used for editing the footage, adding visual effects, and/or other processing. Working with the lower-resolution version of the footage reduces the computing and network resources required to edit the footage and to share the footage among the various parties that are working on the footage. However, the corresponding high-resolution footage is typically required to produce the final version of the television program or movie and matching the lower-resolution footage used during the editing process with the corresponding high-resolution footage can be a time consuming and difficult process. In many cases, decision lists may be supplied as part of the editorial process to assists with cuts, color grading information, and in some cases the process may require manually browsing through large quantities of high-resolution video content to identify the appropriate high-resolution clips of source content. The operator may

struggle to identify appropriate clips from the source content, as many clips of the high-resolution content may include subtle differences that are difficult to identify with the human eye. Hence, there is a need for improved systems and methods that enable a technical solution for indexing, searching, and matching video clips independent of timecode and other metadata.

SUMMARY

(2) An example data processing system according to the disclosure may include a processor and a machine-readable medium storing executable instructions. The instructions when executed, cause the processor to perform operations including obtaining first video footage that includes a plurality of first video frames of a first video clip; obtaining second video footage that includes a plurality of second video frames; analyzing the first video footage to generate a plurality of first fingerprints representing each frame of the plurality of first video frames; analyzing the second video footage to generate a plurality of second fingerprints representing each frame of the plurality of second video frames; comparing the first fingerprints to the second fingerprints generate similarity scores representing how similar each of the plurality of first video frames are to the plurality of second video frames; identifying one or more candidate clip matches for the first video clip based on the similarity scores; selecting a best match for the first video clip in the second video footage from the one or more candidate clip matches for the first video clip; and presenting the best match for the first video clip on a user interface of the data processing system.

(3) An example method implemented in a data processing system for matching video content includes obtaining first video footage that includes a plurality of first video frames of a first video clip; obtaining second video footage that includes a plurality of second video frames; analyzing the first video footage to generate a plurality of first fingerprints representing each frame of the plurality of first video frames; analyzing the second video footage to generate a plurality of second fingerprints representing each frame of the plurality of second video frames; comparing the first fingerprints to the second fingerprints generate similarity scores representing how similar each of the plurality of first video frames are to the plurality of second video frames; identifying one or more candidate clip matches for the first video clip based on the similarity scores; selecting a best match for the first video clip in the second video footage from the one or more candidate clip matches for the first video clip; and presenting the best match for the first video clip on a user interface of the data processing system.

(4) An example machine-readable medium according to the disclosure on which are stored instructions which when executed cause a processor of a programmable device to perform operations of obtaining first video footage that includes a plurality of first video frames of a first video clip; obtaining second video footage that includes a plurality of second video frames; analyzing the first video footage to generate a plurality of first fingerprints representing each frame of the plurality of first video frames; analyzing the second video footage to generate a plurality of second fingerprints representing each frame of the plurality of second video frames; comparing the first fingerprints to the second fingerprints generate similarity scores representing how similar each of the plurality of first video frames are to the plurality of second video frames; identifying one or more candidate clip matches for the first video clip based on the similarity scores; selecting a best match for the first video clip in the second video footage from the one or more candidate clip matches for the first video clip; and presenting the best match for the first video clip on a user interface of the data processing system.

(5) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not limited to implementations that solve any or all disadvantages noted in any part of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The drawing figures depict one or more implementations in accord with the present teachings, by way of example only, not by way of limitation. In the figures, like reference numerals refer to the same or similar elements. Furthermore, it should be understood that the drawings are not necessarily to scale.
- (2) FIG. 1A is an example of an Adaptive Frame-based Clip Match (AFCM) system according to the disclosure.
- (3) FIG. 1B is a diagram showing an example production process and workflow for a video production that may use the AFCM system shown in FIG. 1A.
- (4) FIG. 2 is a diagram showing an example showing correlation of source footage and proxy footage to generate turnover footage for a final release of the video production.
- (5) FIG. 3 is a diagram showing an example of an AFCM fingerprint generation process.
- (6) FIG. 4 is a diagram showing an example of the operations associated with generating a fingerprint of a frame of footage, matching the fingerprint to clips of source footage, and generating metadata and/or reports.
- (7) FIG. 5 is a diagram showing an example process for calculating a fingerprint for a digital image including a coarse-grain and fine-grain elements.
- (8) FIG. 6 is a diagram showing an example process for selecting candidate frames from source footage and matching clips of the source footage based on the candidate frames.
- (9) FIG. 7 is a diagram showing an example of a digital image being processed to generate a fingerprint for the digital image.
- (10) FIG. 8 is a diagram showing an example of an AFCM system and inputs and outputs of the AFCM system.
- (11) FIG. 9 is a diagram showing an example of a folding process that may be used to generate the fine-grain element of the fingerprint.
- (12) FIG. 10 is a diagram showing an example clip candidate selection process.
- (13) FIG. 11 is a diagram showing an example clip matching process.
- (14) FIG. 12 is a diagram showing an example folding process for folding a multidimensional hypercube that may be used by the process of FIG. 9 to generate the fine-grain elements of the fingerprint.
- (15) FIG. 13 is a diagram showing an example fine-grain scoring process.
- (16) FIG. 14 is a diagram showing an example fingerprint selection process.
- (17) FIG. 15 is a diagram showing an example process of matching proxy footage without embedded or side-channel metadata to source footage and computing a cut list for high-resolution deliverable footage.
- (18) FIG. 16 is a diagram that shows how the AFCM system may be utilized in a studio environment.
- (19) FIG. 17 is a diagram that shows a typical editorial relationship between footage to a cut of a scene to a cut of a clip.
- (20) FIG. 18 is a diagram of an example use case in which clips of a video content of a sports broadcast are matched using the AFCM techniques provided herein.
- (21) FIGS. 19A, 19B, 19C, 19D are diagrams showing example fingerprint visualizations.
- (22) FIG. 20 is a diagram of an example process for preprocessing an image.
- (23) FIG. 21 is a diagram of an example process for applying an anti-color filter to an image.
- (24) FIG. 22 is a diagram of an example process for fine-grain fingerprint generation.
- (25) FIG. 23 is a diagram of an example process for fingerprint generation.
- (26) FIG. 24 is a diagram of an example dimension reduction process.

- (27) FIG. 25 is a diagram of an example process for generating a color-sensitive fingerprint.
- (28) FIG. 26 is a diagram of an example process for generating a visual representation of a fingerprint.
- (29) FIG. 27 is a diagram of a process for resizing and cropping a frame of footage.
- (30) FIG. 28 is a flow diagram of an example process for matching proxy footage to source footage.
- (31) FIG. 29 is a diagram showing details of fingerprint generation and using the fingerprints generated to perform a matching operation.
- (32) FIG. 30 is a diagram that shows additional details of the matching operations that may be performed by the matching unit shown in FIG. 29.
- (33) FIG. 31 is a diagram of a process for performing a matching operation.
- (34) FIG. 32 is a diagram of an example process 3201 that may be used to calculate the score matrix in the process shown in FIG. 31.
- (35) FIG. 33 is a diagram of a process that may be used to finding a best clip candidate in the process shown in FIG. 31.
- (36) FIG. 34 is a diagram of an example score matrix.
- (37) FIG. 35 is an example of denormalized probability matrix.
- (38) FIG. 36 shows an example process for identifying clip candidates from the source footage that match the clips in the proxy footage.
- (39) FIG. 37 shows an example matrix in which the matches for the proxy frames and the source frames have been shaded.
- (40) FIG. 38 shows another example matrix in which the matches for the proxy frames and the source frames have been shaded.
- (41) FIG. 39 shows another example matrix in which the matches for the proxy frames and the source frames have been shaded.
- (42) FIG. 40 is a block diagram showing an example software architecture, various portions of which may be used in conjunction with various hardware architectures herein described, which may implement any of the described features.
- (43) FIG. 41 is a block diagram showing components of an example machine configured to read instructions from a machine-readable medium and perform any of the features described herein.

DETAILED DESCRIPTION

- (44) In the following detailed description, numerous specific details are set forth by way of examples in order to provide a thorough understanding of the relevant teachings. However, it should be apparent that the present teachings may be practiced without such details. In other instances, well known methods, procedures, components, and/or circuitry have been described at a relatively high-level, without detail, in order to avoid unnecessarily obscuring aspects of the present teachings.
- (45) Techniques are described herein for matching video content. The techniques provide a technical solution to the technical problem of accurately matching two versions of video footage. Extensive amounts of source footage may be captured for a video production. The source footage is typically high-resolution footage that includes many frames per second. The computing resources required to store and manipulate the source footage is significant even for short segments of video content. Consequently, members of the post-production team of a video production often work with lower-resolution proxy footage that serves as a stand-in for the high-resolution source footage but requires less computing resources to store, transmit, and manipulate than the high-resolution source footage. At some point during the post-production process, the original high-resolution source footage corresponding to the proxy footage must be located and edited to produce the final footage for the video production. However, locating the source footage may be a time consuming and error-prone process. Multiple takes of a shot may have been taken during production, and there may be only subtle differences between the takes. A human user may find it difficult to differentiate

between these versions of the source footage, requiring the user to painstakingly examine the multiple copies of the footage to attempt to identify the correct source footage. Furthermore, the proxy footage may have been manipulated during various stages of the post-production process and it may no longer be obvious to a human user where the footage to create the proxy footage was obtained from the source footage. The matching techniques provided herein solve these and other technical problems by using fingerprinting techniques to generate a frame-level representations of the proxy footage and the source footage. The fingerprints of the source footage and the proxy footage may be compared to identify candidate clips in the source footage that match one or more clips of footage included in the proxy footage. A best-match clip of source content may be identified for each of the clips in the proxy footage. The matching process is automatic, and much faster and more accurate than current editing tools.

(46) Another technical problem solved by the matching techniques provided herein is that the metadata associated with the proxy footage and/or the source footage may be missing, incomplete, or incorrect making it difficult to identify the correct source footage. This metadata may be used to help identify which source footage corresponds to proxy footage. The techniques provided herein may automatically identify and correct such errors in the metadata. These and other technical benefits of the techniques disclosed herein will be evident from the discussion of the example implementations that follow.

(47) FIG. 1A is a diagram that shows an example of an Adaptive Frame-based Clip Match (AFCM) system **100** that may implement the techniques described herein. The AFCM system **100** may be used in television program, movie, or other video production. As will be discussed in greater detail with respect to FIG. 1B, the video production process typically includes multiple stages of production in which source footage **117** is edited to produce finished footage **113** for the production. The edits may include adding visual effects, credits, voice over, color correction, sound effects, and/or other edits that may be made to source footage **117**. The source footage **117** is high-resolution footage that is captured by one or more cameras. The source footage **117** is typically not edited directly. Editing the source footage directly may require significant computing resources, storage resources, and network resources to transfer the source footage **117** among the various participants to the production process. To facilitate the editing process, lower-resolution proxy footage **120** may instead be derived from the source footage **117**. The proxy footage **120** may be used in various stages of the post-production production process and later be exchanged for the corresponding high-resolution source footage **120** when the finished footage **113** is produced. The source footage **117** may be associated with metadata **125** that includes references to the locations in the source footage **120** for the specific footage. The proxy footage **120** may also be associated with metadata **125** that includes references to the content in the proxy footage **120** and references to the corresponding source footage **117**.

(48) The conventional approach to creating the final footage **113** is typically a semi-manual visual process that involves retrieving clips of source footage **117** based on the available metadata leveraging and not limited to industry editorial or color grading tools. However, this process may be slow and error prone, and existing tools may be very slow and cumbersome to use. The metadata for a particular clip of the proxy footage **120** may be missing or incorrect. However, the AFCM system **100** may be configured to analyze proxy footage **120** using the fingerprinting techniques provided herein to identify the source footage **117** associated with a particular clip of proxy footage **120** quickly and accurately. The AFCM system **100** may generate new and/or updated metadata **185** where the metadata is missing or incorrect. Furthermore, the AFCM system **100** may provide various reports, such as but not limited to reports detailing footage clips or shots included in the proxy footage **120** that could not be found in the source footage **117**. The AFCM system **100** may also provide fingerprint visualizations **190** that present a visual representation of the complex fingerprint information generated for frames of proxy footage **120** and/or source footage **117** in a format that allows user to quickly understand the attributes of the frames of

footage. The fingerprints are independent both of resolution of the image and of the encoding scheme of the image frame.

(49) The AFCM system **100** may provide a set of editorial tools for performing various types of edits on footage and/or may be integrated with third-party products as plug-ins or libraries. In some implementations, the AFCM system **100** may provide a set of APIs that enable the third-party editorial tools to access the functionality provided by the AFCM system **100**. In some implementations, the AFCM system **100** may include web-based or cloud-based components that may be accessed by a third-party tool to provide the fingerprinting and matching functionality described herein. Additional details of these and other features of the AFCM system **100** are provided in the examples which follow.

(50) FIG. **1B** is a diagram showing the post-production workflow in which video footage may be captured and processed for a television program, movie or feature film, or other video production in which the AFCM system **100** shown in FIG. **1A** may be used. Activities during the production phase **102** of the workflow may be performed on the set or sets associated with the video production, while activities during the post-production phase **104** may be performed at one or more post-production locations.

(51) A “film shot” or “shot” as used herein refers to a series of frames captured by a camera that runs uninterrupted for a period of time. A “take” as used herein refers to a run of a camera recording a single shot. During filming in the production phase **102**, multiple takes **109** are often captured of a single shot, and multiple cameras **116** may capture each take. The high-resolution source footage **117** captured including the multiple takes captured by the cameras **116** may be sent to one or more post-production facilities during the post-production phase **104**. Final footage **113** for the video production may be created by taking clips from this high-resolution footage, editing the clips, and combining the clips into the final footage **113**. The high-resolution source footage **117** may be stored in the source footage datastore **110**. The high-resolution source footage **117** may also be provided to the transcoding unit **108** to generate the proxy footage **120**. The lower-resolution proxy footage **120** may be used for faster processing of the footage and more efficient transport of the footage among the various entities that are responsible for performing various tasks to create the finished footage **113** for the television program, movie, or other video production. The proxy footage **120** may include a subset of the source footage **117** stored in the source footage datastore **110** and may be used during the post-production phase **104** of development along with timing information and metadata information that links the proxy footage **120** with the source footage **117** stored in the source footage datastore **110**.

(52) Additional content **111** may be created during the post-production phase **104**. The additional content **111** may include but is not limited to visual effects (VFX), credits, and other digital content to be inserted into the video production. The additional content **111** may initially be created using the proxy footage **120** to reduce the computational requirements for manipulating the footage, and the low-resolution versions of the additional content **111** may be added to the proxy footage **120** stored in the proxy footage datastore **107**. In other implementations, the high-resolution source footage **117** may need to be obtained to perform certain types of edits. For example, the VFX may initially be designed and/or produced using clips from the proxy footage **120** but the corresponding clips of high-resolution source footage **117** may need to be obtained to integrate the VFX into finished footage or footage that may be further edited by other participants in the post-production phase **104** to create the finished footage **113**.

(53) Some stages of production phase **102** occur in a sequential manner, such as the filming on set **105** and the production of onset dailies **101** at the end of each day of production **106**. The onset dailies **101** may be raw and unedited footage captured during a day of shooting of the content for the video production. The onset dailies **101** may be viewed by the director, cinematographer, editor, and/or other members of the production crew to review the content that was captured for the day, to identify technical problems in the captured content, to evaluate the performance of actors in the

scenes captured, and to determine whether the scenes were captured by a sufficient number of camera angles. In contrast, many of the stages of post-production phase **104** may be performed at least in part substantially in parallel. However, there may still be some sequential steps in the post-production workflow where delays in the completion of certain tasks may delay the completion of other tasks which may ultimately delay the creation of the finished footage **113**. Domain experts in one stage of post-production may provide their work product to domain experts in the next stage of the post-production phase **104**. Delays in these steps may result in expedited efforts to maintain the post-production and release schedule for the finished footage **113**. However, these expedited efforts may result in increased post-production costs and provide opportunities to introduce errors which may negatively impact the quality of the finished footage **113**.

(54) Errors during the post-production phase **104** may result in the misalignments between the timing information of the proxy footage **120** and the high-resolution source footage **117** may occur during various phases of the post-production phase **104**. These misalignments may introduce errors in the finished footage **113** that may result in expenses and delays and corrections to finished footage **113**. Such misalignments may occur in various stages of the post-production phase **104**, such as but not limited to editing the film to combine shots into sequences of finished footage **113**, color correction to adjust the color, contrast, and exposure of the footage, visual effect (VFX) production, sound editing, score editing, subtitling, ad-placement, and other phases of production. Many stages of the post-production workflow deliver additional audio/video/visual effects (A/V/VFX) content to be inserted into existing footage, to modify existing footage, and/or references existing footage. A robust end-to-end tracking and management system would be required to track all the references to the high-resolution source footage **117** included in the added and/or modified proxy footage **120**. However, each of the stages of the post-production workflow may utilize different equipment and associated software platforms, which may result in discontinuity errors and may introduce additional complexity in tracking all the references to the high-resolution source footage **117** included in the added and/or modified proxy footage **120**. As a result, human intervention is typically required to complete the turnover or handover from one phase to another. Such costly human involvement and other quality control checks can introduce delays and other additional errors and cost. Furthermore, many phases of post-production processes depend on content that has been processed. Thus, delays in one phase of the post-production phase **104** may compound the delays in producing the final footage **113** for a video production.

(55) FIG. 2 is a diagram showing a process in which source footage **117** and the proxy footage **120** derived from the source footage **117** are correlated and matched during the post-production workflow to obtain turnover footage **204**. The turnover footage **204** includes high-resolution footage from the source footage **117** and may be integrated into the final footage **113** or may be further processed before being integrated into the final footage **113**. For example, the turnover footage **204** may be used by a VFX team to add VFX to the high-resolution content to high-resolution content from the source footage **117**. The producer, director or creative editorial team members may initially work with content from the lower-resolution proxy footage **120** to reduce the computing resources required to process the higher-resolution source footage **117**. The turnover footage **204** may then be passed on to another participant in the post-production workflow to further edit the footage or may be utilized for generating the final footage **113** of the video production. The process **200** may be implemented in one or more editorial tools **203**, such as the editorial tools **112** and **114** shown in FIG. 1B. The AFCM system **100** may provide a set of editorial tools for performing various types of edits on footage and/or may be integrated into third-party products as plug-ins or libraries. The AFCM system **100** may provide a set of APIs that permit the third-party products to access the functionality provided by the AFCM system **100**. In some implementations, the AFCM system **100** may include web-based or cloud-based components that may be accessed by a third-party tool to provide the fingerprinting and matching functionality described herein.

(56) The correlation is performed on a portion of the source footage **117** in this example implementation which spans time t.sub.Mo to time t.sub.Mm. The source footage **117** includes a plurality of frames of video content. The source footage **117** includes multiple scenes, including scenes A, B, C, and Z. Scene A is located a time t.sub.M1 **209** in the section **201** of the source footage **117**, where the start of the section the source footage is denoted at t.sub.M0 **207**. The section **201** of the source footage represents the frames of the scene A in this example.

(57) The example shown in FIG. **2** highlights four scenes, but the source footage **117** in other implementations may include a different number of scenes. In this example, the high-resolution versions of the clips may be requested so that they may have VFX added and/or or other processing performed on the clips. The current approach to identifying these clips requires a user to manually browse through the source footage **117** to identify the locations of the desired clips. The user may use timecode information included in the metadata associated with the proxy footage **120** to try to find the clip in the source footage **117**. But this approach is time consuming and error prone. Any modifications that were made to the proxy footage using the editor tools **112** or **114** may result in the timecode references associated with the proxy footage no longer matching the correct timecodes in the source footage **117**. However, the fingerprinting techniques provided herein provide a means for identifying the locations of the clips within the source footage **117** as will be discussed in greater detail in the examples which follow.

(58) The techniques provided herein may provide **1:1** matching of a clip from the proxy footage **120** with a corresponding clip of the source footage **117**. The locations of the clips within the source footage **117** may be represented in the metadata associated with the source footage **117** as frame numbers (also referred to herein as time codes”), which represent the location of the clips within their respective scenes. The clip P.A represented in the proxy footage **120** corresponds to a clip M.A within the scene A of the source footage **117**. The clip may include all or subset of the scene A. The clip P.A is found at time t.sub.P0 **206** of the proxy footage. The clip P.B represented in the proxy footage **120** corresponds to a clip M.B within the scene B of the source footage **117**. The clip may include all or subset of the scene B. The clip P.C represented in the proxy footage **120** corresponds to a clip M.C within the scene C of the source footage **117**. The clip may include all or subset of the scene C. The clip P.D represented in the proxy footage **120** does not have a corresponding clip M.D within the source footage **117** in this example. The clip P.Z represented in the proxy footage **120** corresponds to a clip M.Z within the scene Z of the source footage **117**. The clip may include all or subset of the scene Z. The source footage **117** may be associated with metadata, and the location of each of the scenes within the source footage **117** may be indicated using timecodes that represent the frames of the source footage **117** that make up each scene. The turnover footage **204** includes the high-resolution content from the source footage **117** that corresponds to the selected clips.

(59) The selected clips represented by P.A, P.B, P.C, P.D, and P.Z in this example may be selected by a user. The corresponding clips M.A, M.B, M.C, and M.Z in the source footage **117** are being sought to be included in the turnover footage **205**. The clips M.A, M.B, M.C, and M.Z included in the turnover footage **204** may be further processed in the post-production phase **104** or may be used as is in the final footage. The user may select the clips P.A, P.B, P.C, P.D, and P.Z using one of the editing tools **112** or **114** by reviewing the proxy footage **120** to identify the desired clips. The AFCM system **100** may generate a fingerprint or fingerprints for each of the desired clips of the proxy footage **120** and the fingerprints generated from the proxy footage **120** may be compared to fingerprints generated from the source footage **117** by the AFCM system **100**. If a match is found for a desired clip, the high-resolution footage may be extracted from the source footage **117** and the included in the turnover footage **204**. If the desired clip is not found, such as clip M.D in the example of FIG. **2**, the clip's source may not have been included and may need to be found elsewhere. Furthermore, in some instances, the proxy footage **120** associated with the clip may have been modified by VFX or other processing which caused the fingerprint or fingerprints of the

desired clip to no longer match the fingerprint of the corresponding clip in the source footage **117**. The AFCM system **100** may identify any clips could not be found in a report of the reports **195** generated by the AFCM system **100**. The unmatched clips, such as clip M.D, may need to be manually located in the source footage. The unmatched clips may be missing or mislabeled in the metadata associated with the source footage **117**.

(60) FIG. **3** is diagram that shows one example of a high-level AFCM fingerprint computation process that may be implemented by the AFCM system **100**. The process shown in FIG. **3** may be performed on high-resolution footage files **302**, which may be obtained from the high-resolution source footage **117** stored in the source footage datastore **110**. The fingerprinting process may be performed as the footage files **302** are stored in the source footage datastore **110**, may be performed on demand as a search is conducted for high-resolution footage **117** stored in the source footage datastore **110**, or may be scheduled to be performed periodically for high-resolution source footage **117** that has been added to the source footage datastore **110**. In the example shown in FIG. **3**, the footage files **302** comprise N frames of footage, where in N is an integer greater than 1. The footage files **302** may include one or more files of footage. Each footage file of the footage files **302** includes X frames of the high-resolution footage **117**, where X is an integer greater than 0. Each frame of the footage files **302** may be processed by the fingerprint generation unit **303** and the fingerprint information associated with the frames of footage may be stored in a fingerprint indexed datastore **304**. The process used by the fingerprint generation unit **303** for generating the fingerprints is described in detail in the examples which follow. Furthermore, the fingerprint indexed datastore **304** is also described in detail in the examples which follow. The fingerprint indexed datastore **304** is configured to facilitate searching for frames of footage that have a fingerprint that matches that of a candidate frame of proxy footage. In some implementations, the fingerprint indexed datastore **304** may be stored in the source footage datastore **110**. In other implementations, the fingerprint indexed datastore **304** may be a separate datastore.

(61) FIG. **4** is a diagram that shows an example stack of processes that may be used by the fingerprinting techniques of the AFCM system **100** to match a candidate clip from the proxy footage **120** with a corresponding clip in the high-resolution source footage **117**. The matching is done by generating one or more fingerprints for the candidate clip and matching the one or more fingerprints to the fingerprints of the high-resolution source footage **117**. As discussed with respect to FIG. **3**, the source footage **117** stored in the source footage datastore **110** may be analyzed by the fingerprint generation unit **303** and stored in the fingerprint indexed datastore **304**.

(62) In the frame fingerprint operation **402**, each frame of digital content is processed to generate a fingerprint the frame. The fingerprint of the frame may be used to search for frames in the source footage datastore **110**. The fingerprint may include a coarse-grain fingerprint component and a fine-grain fingerprint component. The details of how the fingerprint is calculated are described in greater detail in the examples which follow.

(63) In the frame candidate selection operation **403**, a list of frames of the source footage **117** of the source footage datastore **110** may be determined by comparing the coarse-grain and fine-grain fingerprints with the fingerprint information stored in the fingerprint indexed datastore **304**. Potential matches to the reference frame of the proxy footage **120** are provided. The system may automatically select a best match or may present the candidates to a user via user interface configured to present the candidate frames and the user may provide an input selecting a candidate frame.

(64) In the clip candidate selection operation **404**, the frame candidates selected in operation **403** are used to generate clip candidates that may match the candidate clip from the proxy footage **120**. In the clip match selection operation **405**, a best match for the candidate clip may be selected from among the clip candidates if multiple clip candidates are found.

(65) In the user interface and application stage **406**, the results of the matching may be presented to the user via a user interface of the system. The system may also generate an edit decision list

(EDL) file used in post-production of film and video editing, an advanced authoring format (AAF) file used to exchange digital media and metadata between platforms, and/or other file formats. The system may also generate one or more reports. For example, the system may generate a report for reference frames of the proxy footage **120** for which a match could not be found.

(66) FIG. 5 is a diagram showing an example process **500** for generating the fingerprint from a frame of video content. The frame of the video content may be a digital image **504** of any size. The aspect ratio and size of the frame may depend upon the format of the video content from which the frame of video content has been extracted. The process may be generated by a fingerprint generation unit of the AFCM system **100**.

(67) The fingerprint process may include an image preprocessing operation **503**. The image preprocessing operation **503** may perform various operations on the digital image **504**, such as but not limited to converting the image **503** to a grayscale image. The fingerprint generation unit may generate a grayscale image from a color image by converting color values in the image to luminance values. The pre-processing operation **503** may convert a color image to grayscale to reduce errors that may have been introduced by color compression or color correction that have been applied during editing. The pre-processing operation **503** may include applying a Gaussian blur to the digital image **504** to reduce compression errors. The pre-processing operation **503** may include a cropping operation for selecting a portion of the digital image **504** that should be consistent across different aspect ratios. The pre-processing operation **503** may include converting from pixels to a grid of histograms of oriented gradients to decrease sensitivity to minor zooms, translations, and/or color grading changes made during editing. The pre-processing operation **503** may apply one or more of these operations to the digital image **504** and/or other such operations to prepare the image for analysis by the coarse-grain fingerprinting pipeline **501** and the fine-grain fingerprinting pipeline **502**. The pre-processed image may then be provided to a coarse-grain fingerprinting pipeline **501** and a fine-grain fingerprinting pipeline **502**.

(68) The coarse-grain fingerprinting pipeline may be configured to generate a coarse-grain fingerprint for the image **504**. The coarse-grain fingerprinting pipeline **501** may include a resizing operation **510** in which the preprocessed image may be resized to a first smaller standard size. In the example shown in FIG. 5, the resizing operation **510** resizes the preprocessed image to 64×64 W pixels. However, other implementations may resize the preprocessed image to a different standard size that is equal to or smaller than the pre-processed image generated by the pre-processing operation **503**. The term “W” refers to the quantized size of each image pixel in bits (the quantization need not be linear). Each pixel may be represented by one or more bits. For example, each pixel may be represented by 4 bits in some implementations and by 64 or more bits in other implementations. The resized image may be flattened into a vector of 4096 dimensions.

(69) The coarse-grain fingerprinting pipeline **501** may include a locality-sensitive hashing (LSH) operation **511** in which a hash of the resized image or the flattened vector representing the resized image may be created. The LSH operation similarly reduces the dimensionality of high-dimensional input data to a lower-dimensional version while preserving relative distances between items. The hash provides a lower-dimensional representation of the resized image (also referred to herein as a “coarse-grain fingerprint”) that may be provided as an input to the folding matching operation **513**. In some implementations, a random projections LSH operation may be used, while other LSH techniques may be used in other implementations.

(70) The fine-grain fingerprinting pipeline **502** may be configured to generate a fine-grain fingerprint for the image **504**. The fine-grain fingerprinting pipeline **502** may include a resizing operation **509** in which the preprocessed image may be resized to a second smaller standard size larger than the first standard size used by the coarse-grain fingerprinting pipeline **501**. In the example shown in FIG. 5, the resizing operation **509** resizes the preprocessed image to 256×256 W pixels. The term “W” refers to the quantized size of each image pixel in bits, and each pixel may be represented by one or more bits. However, other implementations may resize the preprocessed

image to a different standard size that is smaller than the pre-processed image generated by the pre-processing operation **503** but larger than the first standard size used by the coarse-grain fingerprinting pipeline **501**. Furthermore, different quantization sizes may be used in different implementations.

(71) The fine-grain fingerprinting pipeline **502** may include a grid division operation **505** in which the resized image from the resizing operation **509** may be divided into a set of smaller sized cells. In the implementation shown in FIG. 5, the resized image is divided into an 8×8 grid of 32×32 W sized cells **506** for a total of 64 cells. Other implementations may be subdivided into a different number of cells of a different size than that shown in the example of FIG. 5. The size and number of cells may depend, at least in part, on the size of the resized image produced by the resizing operation **505**.

(72) The fine-grain fingerprinting pipeline **502** may include a folding LSH operation **507** in which the cells **506** may be analyzed and converted into a bit string **508**. Additional details of the folding LSH operation are shown in FIG. 9 which is described in detail the examples which follow. In this example, each of the 64 cells **506** cells may be analyzed to generate a 128-bit string, and these strings may be combined to generate an 8192-bit string. The bit string **508** represents the fine-grain fingerprint. The fine-grain fingerprint may be appended or otherwise combined with the coarse-grain fingerprint to create the fingerprint **513** for the image **504**.

(73) FIG. 6 is a diagram of an example clip-matching process **600**. The clip-matching process may be implemented by the AFCM system **100**. The clip-matching process may use the fingerprint information generated using the process shown in FIG. 5. In this example, the proxy footage **120** includes two clips A' and B' for which the corresponding clips from scenes A and B in the high-resolution footage source footage **117** are desired. The clip-matching process attempts to find these two clips in the source footage **117** by matching the fingerprints of the desired claims with the fingerprints stored in the fingerprint indexed datastore **304**. The proxy footage **120** may be the proxy footage **120** of the preceding examples. The section **601** of the source footage **117** represents the frames of the scene A in this example. The clip A' may include all or a subset of the frames of the scene A.

(74) The fingerprint generation operation **603** may generate the fingerprints of frames of the source footage **117** that includes the clips of interest. The fingerprinting operation **603** may use the process shown in FIG. 5 to generate the fingerprints for the source footage **117** and store the fingerprint information in the fingerprint indexed datastore **304**. As discussed in the preceding examples, the fingerprint information for the source footage **117** may be generated on demand when a particular clip or clips of the source footage **117** are to be retrieved. In other implementations, the fingerprint information for the source footage **117** may be generated in advance and stored in the fingerprint indexed datastore **304**. In implementations where the fingerprints of the source footage **117** are generated on demand, the proxy footage **120** may include reference information that identifies the source footage **117** associated with the desired clips from the proxy footage **120**. As discussed in the preceding examples, these references may be timecodes that reference a specific location with the source footage **117**. The references may also include additional information, such as a storage location and/or filename or filenames of the source footage **117** from which the proxy footage **120** was derived.

(75) The clip candidate and matching operation **607** includes identifying candidate clips from the source footage **117** and selecting a best match from the candidate clips. The clip candidate and matching operation **607** may include a fingerprint generation operation **608** in which the fingerprint or fingerprints for the clips in the proxy footage **120** using the fingerprinting process shown in FIG. 5. The frame candidate selection process **605** uses the fingerprint or fingerprints to identify candidate frames in the fingerprint indexed datastore **304**. In some implementations, the frame candidate selection process **605** may include splitting the coarse-grained fingerprint generated by the coarse-grain fingerprinting pipeline **501**. The frame candidate selection process **605** may split

the coarse-grained fingerprint into N partitions, where N is an integer value greater than 1. The frame candidate selection process **605** may then query the fingerprint indexed datastore **304** to identify frames that contain any of the N partitions. These frames may then be filtered and scored by the filtering and scoring fingerprint matching process **606**. Additional details of the filtering and scoring fingerprint matching process **606** are shown in FIG. **14**, which is described in detail in the examples which follow. The frame candidates are then provided to the clip candidate and matching process **607** to select clip candidates and to determine a clip that best matches the clip from the proxy footage **120**. FIG. **10** provides an example implementation of a clip candidate selection process, and FIG. **11** provides an example implementation of a clip matching process. The examples shown in FIGS. **10** and **11** are discussed in detail below. This process shown in FIG. **6** may be repeated for each of the clips in the proxy footage **120**.

(76) FIG. **7** is a diagram that illustrates an example of the fingerprinting process shown in FIG. **5** being applied to a color digital image frame **702**. The color image frame **702** may be a frame of the source footage **117** or the proxy footage **120**. The color image frame **702** may be a Red Green Blue (RGB) color image or may use a different color model to represent the color in the image frame **702**. The techniques provided herein are not limited to source images that use a specific color model or set of color models. The color image frame **702** is converted to a grayscale image **703**. As discussed in FIG. **5**, the preprocessing operation **503** may convert the color image frame **702** to the grayscale image **703** and may also perform other preprocessing operations on the image. The image sizes for the image frame **702** and the grayscale image **703** may be selected based on the particular implementation and the desired output. Furthermore, other settings such as the grid size, the folding cell sizes, and/or other parameters may also be configured based on the particular implementation and the desired input. These and other parameters may be adjusted to trade off for accuracy versus speed. Less detailed fingerprints may be generated to speed up the search process but may result in the reduced accuracy of the matches obtained. In some implementations, the system may provide a user interface that allows the user to configure default settings to be used for fingerprinting generation. In some implementations, the system may also provide options for defining the fingerprint generation process on a user interface in which the user may conduct a search for frames of high-resolution source footage **117** that includes a particular frame.

(77) FIG. **8** is a diagram that shows an example of some types of inputs that the AFCM system **801** may receive and the types of outputs that the AFCM system **801** may output. As discussed above, the AFCM system **801** may receive one or more source files **803** that include high-resolution source footage **117** as an input. The AFCM system **801** may receive one or more proxy files **802** that include proxy footage **120** as another input. The AFCM system **801** may optionally receive EDL files, AAF files, and/or other source files formats that include metadata for the source files **803**. The metadata may include timecode references that identify the location of specific scenes, clips, color lookup tables (LUTs), editorial and color grading effects, VFX, or other content in the source files **803**. The AFCM database **804** shown in FIG. **8** may be implemented by the source footage datastore **110** and/or the fingerprint indexed datastore **304**. The fingerprints generated by the AFCM system **801** and stored in the AFCM database **804** are separate from the metadata.

(78) The AFCM system **801** may do metadata repair and/or metadata generation. The metadata may include but is not limited to a cut list of timecodes (hour, minutes, second, and frame) that indicate where cuts are located within the source footage **117**. The metadata may be missing or non-existent for a particular source file or source files. The AFCM system **801** may analyze the source file or source files to generate the missing metadata. The newly generated metadata **806** may include the start and end times of cuts included in the source footage **117**.

(79) The AFCM system **801** may output an AFCM report **805** for requested cuts for which a match could not be found. The one or more source files **803** of the source footage **117** may be missing the requested cut or the requested cut may be incorrectly indexed. The report may indicate that a requested clip, missing zoom, or other request content is missing or incorrectly indexed. This

allows problems to be quickly identified and addressed, thereby providing improved quality control in the post-production workflow.

(80) FIG. 9 is a diagram that shows an example folding process 900 that may be used to implement the folding LSH operation 507 performed by the fine-grain fingerprinting pipeline 502 shown in FIG. 5. The folding technique shown in FIG. 9 may be performed on a set of cells, such as the cells 506, derived from the resized image created by the resizing operation 509 of the fine-grain fingerprinting pipeline 502.

(81) The folding LSH operation shown in FIG. 9 are folded two times by folding operation 902. The folding operation 902 may be implemented by the folding process shown in FIG. 12, which is described in detail in the examples which follow. The output from the folding operation 902 is provided as an input to the LSH operation 906, and the LSH operation 906 outputs a first 32-bit hash of the hashes 907.

(82) The output of the folding operation 902 is provided to the folding operation 903 as an input. The folding operation 903 performs one folding operation on the input. The folding operation may also be implemented using the technique shown in FIG. 12. The output of the folding operation 903 may be input to an LSH operation 906 to obtain a second 32-bit hash of the hashes 907.

(83) The output of the folding operation 903 is provided to the folding operation 904 as an input. The folding operation 904 also performs one folding operation on the input. The folding operation may also be implemented using the technique shown in FIG. 12. The output of the folding operation 904 may be input to an LSH operation 906 to obtain a third 32-bit hash of the hashes 907.

(84) The output of the folding operation 904 is provided to the folding operation 905 as an input. The folding operation 905 also performs one folding operation on the input. The folding operation may also be implemented using the technique shown in FIG. 12. The output of the folding operation 904 may be input to an LSH operation 906 to obtain a fourth 32-bit hash of the hashes 907.

(85) The first, second, third, and fourth hashes may be combined together to form the fine-grain cell fingerprint, which is passed as a parameter to the folding matching operation 908, which may implement the folding matching operation 513 shown in FIG. 5. While the example shown in FIG. 9 includes 64 32×32 cells as the input and outputs 32-bit hashes from each of the folding operations to produce a 128-bit fine-grain cell fingerprint, other implementations may receive a different number of cells of a different size and may be configured to output a different number of bits from each of the LSH operations. Thus, the fine-grain fingerprint may be a different length in other implementations.

(86) FIG. 12 is a diagram that shows an example of the folding techniques that may be used to generate a fingerprint according to the techniques provided herein. The folding technique shown in FIG. 12 may be used to implement the folding operations 902, 903, 904, and 905 shown in FIG. 9. The folding technique shown in FIG. 12 may be used to determine a portion of the fine-grain cell fingerprint.

(87) The possible space for all image vectors may be thought of as a hypercube. In the example shown in FIG. 12, this space is represented by a 1024-dimension hypercube 1202. The cell image vector 1201 may be folded by taking the absolute value of the values included in the vector to generate the folded vector 1203 so that all the values fall into the positive orthant. The values in the folded vector 1203 may then have 0.5 subtracted from them to shift the possibility space of the vector 1204 to be distributed around the origin. This folding process compresses the values used to generate the fingerprint into a smaller space. As a result, differences between similar frames of content are increased and differences between frames of content that are less similar are decreased. As a result, the results produced using the matching techniques described herein may provide more accurate results. Many frames of the source footage 117 may be very similar. The source footage 117 may include multiple shots of a scene and/or other footage that may appear very similar. By

emphasizing the differences between similar frames, the AFCM system may find candidate clips that more accurately match those of the proxy footage. The folding process shown in FIG. 12 allows the AFCM to express the difference between very similar frames of content using a limited number of bits to represent the content of the frames.

(88) FIG. 10 is a diagram showing an example of a clip candidate selection process 1007 which may be used to identify clips in the source footage 117 that are similar to clips in the proxy footage 120. One or more clip candidates from the source footage 117 may be identified.

(89) The source footage 117 may be accessed in blocks of frames, referred to herein as frame pages, that include a plurality of frames of source footage 117. In some implementations, the fingerprints associated with the plurality of frames of source footage 117 may be accessed from the fingerprint indexed datastore 304 rather than obtaining the frames of the source footage. In implementations where the frames of the source footage 117, the fingerprints for at least a portion of the source footage 117 may be generated using the fingerprint generation techniques discussed in the examples provided herein and stored in the fingerprint indexed datastore 304. The fingerprints of the frames of the proxy footage 120 may also be generated if they have not yet been generated.

(90) The process 1007 may include an operation of finding the start and end of the clip candidate(s) 1050 in the source footage 117 frames or fingerprint data representing the frame. The length of the clip of the proxy footage 120 should typically match the length of the clip in the source footage 117. Once a candidate frame has been found that is similar to the start or end frame of the clip of the proxy footage, the frame at the other end of the clip should be found at the respective number of frames from the candidate frame that is similar to the start or end frame of the clip of the proxy footage.

(91) As indicated above, the clip being sought in the source footage 117 may be a subset of the frame page 1003 or a superset of the frame page 1003. The first frame 1002 of the page 1003 is denoted by “s” which represents the initial estimated of the start of the clip. The last frame 1004 of the frame page 1003 is denoted by “e” on the diagram. The search may include additional frame pages, such as the frame page 1009 which has a last frame 1010 denoted by “e” on the diagram.

(92) The search for the start and end frames of the clip candidate(s) 1050 may begin with searching for the start frame of the clip at the first frame 1002 of the frame page 1003 and for the end frame of the clip at the last frame 1004 of the frame page 1003. The fingerprint of the start frame from the proxy footage 120 may be compared to each candidate frame to determine a similarity score and the candidate frame may be selected as a start frame of a clip candidate if the similarity score meets a predetermined threshold. Similarly, the fingerprint of the end frame from the proxy footage 120 may be compared to each candidate frame to determine a similarity score and the candidate frame may be selected as an end frame of a clip candidate if the similarity score meets a predetermined threshold.

(93) A candidate clip may be identified for each pair of start and end frames that satisfy the similarity score requirements. The candidate clips may be identified by conducting a binary search of the frames of the frame page being searched. The candidate clip may be added to the clip candidates 1050 for additional processing to identify a best match for the clip. There may be multiple clips that appear to be quite similar, and thus multiple clips may need to be considered when selecting which clip to be used from the source footage 117. The clip candidate information may include the information that identifies the location of each candidate clip in the source footage, such as timecode information, filename information, and/or other information that may be used to locate and obtain the high-resolution footage of the clip from the source footage 117.

(94) FIG. 11 is an example clip match selection process 1100. The process 1100 may implement, at least in part, the clip candidate and matching operation 607 shown in FIG. 6 or for selecting a best match clip from the clip candidates 1050 determined in the process 1007 shown in FIG. 10. In operation 1102, the source clip candidates 1101 are analyzed and a clip score and confidence interval are calculated for each clip. The clip score and confidence level are based on the average

score of the matched frames within the clip. In operation **1105**, a confidence interval filter is applied and any clips whose confidence interval does not overlap the confidence interval of the lowest (i.e., the best) clip score is discarded. If there is only one clip remaining to be processed, the remaining clip is selected as the best match clip **1107**. Otherwise, in operation **1103** more samples are obtained, and one or more frame from each clip candidate **1101** is scored against a proxy clip **1104**.

(95) FIG. **13** is a diagram that shows an example of a fine-grain scoring process **1300**. The fine-grain scoring process **1300** may be used by the frame candidate selection process **605** shown in FIG. **6** when selecting candidate frames from the fingerprint indexed datastore **304**. In the example shown in FIG. **6**, the fine-grain scoring process **1300** compares Image A from the source footage **117** with Image B from the proxy footage **120**. Image A and image B have been divided into cells such as those generated by the grid division operation **505** of the fine-grain fingerprinting pipeline **502**.

(96) The process **1300** calculates the maximum error of all cell errors, wherein 'w' is a weight vector of constants, and 'HD' represents the Hamming Distance, the maximum error is calculated by calculating the error of each cell by comparing cell A_{ij} of the grid **1301** with cell B_{ij} of the grid **1302** for each value of i and j, where 'i' represents a column and 'j' represents a row of the grids **1301** and **1302**: $Ve = HD(Fold_2_LSH, Fold_2_LSH), HD(Fold_3_LSH, Fold_3_LSH), HD(Fold_4_LSH, Fold_4_LSH), \text{ and } HD(Fold_5_LSH, Fold_5_LSH)$. Cell Error = $(w \cdot Math.Ve)$

(97) The weight vector may, for example, include a plurality of numerical weight values. An example weight vector w may be [1, 0.5, 0.25, 0.1]. The specific weights applied to each of the elements may vary depending upon the specific implementation.

(98) The error values for each pair of cells are computed based on the values produced by the first, second, third, and fourth fold operations performed on the respective cell values from the grids **1301** and **1302** and calculating the Hamming Distance between values from the grid **1301** and the grid **1302**. The Cell Error for each cell is calculated by taking the dot product of the vector of error values Ve calculated for each pair of cells and the weight vector. The maximum Cell Error value may then be determined for the two images.

(99) FIG. **14** is a flow diagram of an example process **1400** that the AFCM system may use to generate and prune a set of fingerprint candidates based on similarity scores. The similarity score in process **1400** includes a coarse-grain score and a fine-grain score component. The process **1400** may be used to implement, at least in part, the filtering and scoring fingerprint matching process **606** shown in FIG. **6**.

(100) The process **1400** uses the following threshold values when selecting fingerprint candidates: Adaptive.Coarse.Threshold = $\min(\minCandidateCoarseScore * 1.5 + 3, C1)$

Adaptive.Fine.Threshold = $\min(\minCandidateFineScore * 1.5 + 3, C2)$

where C1 and C2 are threshold constants, minCandidateCoarseScore refers to the minimum coarse score from all candidates, and the minCandidateFineScore refers to the minimum fine score from all candidates. The multipliers applied to the minCandidateCoarseScore and the minCandidateFineScore and the constants C1 and C2 are examples of one implementation. Other multipliers and constant values may be used in other implementations.

(101) The process **1400** may include an operation **1401** of accessing image fingerprint candidates. The image fingerprint candidates may be obtained from the fingerprint indexed datastore **304**.

(102) The process **1400** may include an operation **1402** in which the coarse-grain score of the candidate is compared to the adaptive coarse threshold value. If the coarse-grain score of the candidate is greater than or equal to the adaptive coarse threshold value, then the process continues with operation **1404** in which the image fingerprint candidate is discarded. Otherwise, if the coarse-grain score of the candidate is less than the adaptive coarse threshold value, the process **1400** may continue with operation **1403**.

(103) In operation **1403**, the fine-grain score is compared to the adaptive fine threshold value. If the

fine-grain score of the candidate is greater than or equal to the adaptive fine threshold value, then the process continues with operation **1404** in which the image fingerprint candidate is discarded. Otherwise, if the fine-grain score of the candidate is less than the adaptive fine threshold value, the process **1400** may continue with operation **1406** in which the candidate fingerprint is selected.

(104) FIG. **15** is an example use case in which the AFCM system performs matching with proxy footage without any embedded or side-channel metadata to the corresponding source footage, and the AFCM system **100** computes the metadata for the for the high-resolution deliverables. The proxy footage **120** may not have metadata or timecode information attached or available. However, the AFCM system **100** may be configured to use the fingerprint determination processes discussed in the preceding examples to correlate the proxy footage **120** with the source footage **117**. The AFCM system **100** uses the fingerprint matches to create new metadata cut list and effect list **1504** for the proxy footage **120**. The AFCM system **100** can generate a report **1505** that identifies clips that from the proxy footage **120** for which a matching clip in the source footage **117** could not be found. The metadata cut list and effect list **1504** may be provided to editorial tools **114** to complete the post-production workflow. The finished footage **113** may be generated by extracting the clips from the source footage **117** that match those included in the proxy footage **120**.

(105) FIG. **16** is a diagram that shows how the AFCM system may be utilized in a studio environment. The AFCM system **100** may include a user application and an AFCM engine. The user application may provide a user interface **1608** that allows the user to select proxy footage from the proxy footage **120** for which the corresponding clips of high-resolution source footage **117** of the source footage **117** are desired. The user interface may also provide a means for the user to select metadata **125** to be corrected by the AFCM system **100** using the fingerprinting and matching techniques provided herein and/or to generate new metadata for source footage **117** for which the metadata is unavailable or inaccurate. The matching technique may use the fingerprint information stored in the database **1603**.

(106) The metadata and the source footage **117** output by the AFCM system **100** may be utilized by video production editorial tools to add VFX, credits, color correction, and/or other actions on the post-production workflow. As discussed in the preceding examples, the AFCM system **100** may be configured to generate a report that identify clips included in the proxy footage **120** that could not be matched in the source footage **117** as well as other relevant information regarding the source and proxy files and footage. The report may be provided to a quality control team associated with a video production. The quality control team may be able to manually identify mislabeled proxy content and/or to notify the post-production team that certain requested content is missing. Alternative clips may also be selected if the missing clips cannot be located. The AFCM system **100** may also output a metadata cut list and effect list which may be utilized by editorial tools **114**.

(107) The AFCM system **100** may be configured to operate in a batch mode in which a user may select a batch of content to be processed offline by the AFCM system **100**. The user may submit the batch of content to be processed. The batch of content to be processed may include footage for which the metadata is to be repaired or created. The batch of content may also include proxy footage **120** for which clips of high-resolution source footage **117** are being requested. The batch processing allows the user to schedule tasks to be performed by the AFCM system **100** without requiring the user to monitor the progress of the tasks. The AFCM system **100** may notify the user when the batch of content has been processed. A report may be generated upon completion of the batch indicating those requests that completed successfully and those requiring user intervention.

(108) FIG. **17** is a diagram that shows a typical editorial relationship between footage to a cut of a scene to a cut of a clip. The source footage **117** shown in FIG. **17** includes N frames of footage including frames F.sub.1 **1702** to frame F.sub.N **1707**. The source footage **117** includes one or more scenes, such as the scene **1703**, captured by a camera or generated via computer graphics or other visual effects tools. The scene **1703** includes frames F.sub.4 **1706** to frame F.sub.12 **1708** in this example. During the editorial process, portions of or all the frames of the scene **1703** may be

selected (or cut) as a clip **1704** for the purposes of creating final footage **113**. In the example shown in FIG. **17**, the clip **1704** includes frames F.sub.6 **1705** to frame F.sub.11 **1709**. The specific number of frames that may be included in a scene or a clip may vary.

(109) FIG. **18** is a diagram of an example use case in which clips of a video content of a sports broadcast are matched using the techniques described herein. In this example, the proxy footage **120** includes four clips **1802**, **1806**, **1807**, and **1808** for which the corresponding clips of high-resolution footage **117** are being sought. The AFCM system **100** performs the fingerprinting and matching operations discussed in the preceding examples to find the corresponding clips from the source footage **117**. In this example, the AFCM system **100** has identified the high-resolution clips **1809**, **1810**, **1811**, and **1812**. In this example implementation, the four clips **1802**, **1806**, **1807**, and **1808** may include a few seconds of highlights from a sports broadcast but the AFCM system **100** may use the clips to identify longer, high-resolution version of proxy clips.

(110) FIG. **20** is a diagram of an example process **2001** for preprocessing an image. The process **2001** may be used to implement the image preprocessing operation **503** shown in FIG. **5** for preprocessing an image frame for the coarse-grain fingerprinting pipeline **501** and the fine-grain fingerprinting pipeline **502**. The input to the process **2001** is an image frame **2007** of any size or resolution.

(111) The process **2001** may include an operation **2002** of converting the image frame **2007** to grayscale. Converting the image frame **2007** to grayscale from a color image may reduce errors introduced by color compression or color correction made to the frame.

(112) The process **2001** may include an operation **2003** of resizing and cropping for tall and wide images. The grayscale image output by operation **2002** may be cropped and resize twice: a first version having a taller aspect ratio and a second version having a wider aspect ratio. The first version of the image may have an aspect ratio where the height of the image is greater than the width of the image, such as but not limited to a 3:4 aspect ratio. The second version of the image may have an aspect ratio where the width of the image is greater than the height of the image, such as but not limited to a 16:9 aspect ratio. However, the first version of the image may have an aspect ratio where the width of the image is greater than the height of the image. For example, the first version of the image may have an aspect ratio of 4:3 and the second version of the image may have an aspect ratio of 2:1. The aspect ratios discussed herein are merely examples that illustrate the concepts discussed herein do not limit the techniques in the instant application to these specific aspect ratios. The first and second version of the image are processed separately for each of the subsequent operations of the process **2001**. The resizing and cropping operation may be performed by the process **2701** shown in FIG. **27**.

(113) The process **2001** may include an operation **2004** in which an anti-color filter is applied to the first and second images output by the operation **2004**. An example of an anti-color filter that may be used to implement operation **2004** is shown in FIG. **21**.

(114) The process **2001** may include an operation **2005** of flattening the first and second images processed by the anti-color filter in operation **2004**. The first image may be flattened from a two-dimensional vector into a first one-dimensional vector and the second image may be flattened from a two-dimensional vector into a second one-dimensional vector.

(115) The process **2001** may include an operation **2006** of shuffling the elements of the vectors produced by the operation **2005**. The elements of the vectors are shuffled using a predetermined shuffling routine so that the same reordering occurs each time that the operation **2006** is performed on a vector. The shuffling of the elements ensures that any correlation between pixel adjacency and vector element adjacency is eliminated. The shuffled vectors may be provided as an output **2008** from the process **2001**.

(116) FIG. **21** is a diagram of an example process **2101** for applying an anti-color filter to an image. The process **2101** may implement the operation **2004** of the process **2001** shown in FIG. **20** in which the process **2101** may be separately applied to the first and second images.

(117) The input **2107** to the process **2101** may be an image frame of footage or other digital image. The image frame may have been preprocessed to grayscale, resized, cropped, and/or otherwise processed prior to the process **2101** being applied.

(118) The process **2101** may include an operation **2102** of convolving the input image **2107** with a kernel. The kernel may be a difference of Gaussians (DoG) kernel in some implementations. Other implementations may utilize other filters that may serve a similar purpose as the DoG kernel. The DoG kernel involves subtracting of a first Gaussian blurred version of the input image from a second Gaussian blurred version of the input image, where the second image is less blurred than the first image. Gaussian blurring may be used to suppress high-frequency spatial information and subtracting the first image from the second image may preserve the spatial information that falls within the range of frequencies which are preserved in the first blurred image and the second blurred image. Thus, the DOG may serve as a spatial band-pass filter that may attenuate frequencies in the original image which are farther from the band center. In some implementations, the sigma values for the DOG kernel may be $\sigma_{\text{sub.1}}=1$ and $\sigma_{\text{sub.2}}=10$. Other sigma values may be used in other implementations.

(119) The process **2101** may include an operation **2103** of subtracting a mean pixel value excluding outliers. In operation **2103**, the mean pixel value excluding the N largest and smallest values is calculated and subtracted from each of the pixels of the image output by operation **2102**. In one implementation, the value of N may be 1000, which causes the 1000 largest and 1000 smallest values to be excluded from the calculation of the mean. Other values of N may be selected depending on the degree of clipping that exists in the color graded footage.

(120) The process **2101** may include an operation **2104** of calculating a pixel variation value for the image output by operation **2103** and dividing the pixel values of the image by the pixel variation value. The pixel variation may be calculated using a technique such as mean absolute deviation (MAD) and normalized to effectively normalize variation. The MAD represents the average distance between each pixel value of the image and the mean pixel value. A minimum threshold may also be used to avoid divide by zero. The mean absolute deviation can be calculated by calculating the distance of each pixel from the mean pixel value excluding outliers obtained from operation **2103** and taking the average.

(121) The process **2101** may include an operation **2105** of multiplying the pixel values resulting from the operation **2104** by a constant. The optimal value of the constant C may depend upon the convolution kernel that is being used in operation **2102** and the outlier trimming occurring in operation **2103**. For example, the constant values C in one implementation is 5 based on these factors. However, other implementations may use a different constant value. The process **2101** may include an operation **2106** of applying a sigmoid function to the resulting pixel values from operation **2105**. The results of operation **2106** may be provided as an output **2108** of the process **2101**.

(122) FIG. 22 is a diagram of a process **2201** for fine-grained fingerprint generation. The process **2201** may be implemented as an alternative to the fine-grain fingerprinting pipeline **502** of the fingerprint generation process shown in FIG. 5. The output **2008** of the image preprocessing process **2001** may be provided as an input **2208** of the process **2201**. The output **2008** of the image preprocessing process **2001** comprises vectors as discussed above with respect to FIG. 20.

(123) The process **2201** may include an operation **2202** of combining dimensions of a vector representing an image to produce a lower dimension vector. In one implementation, the number of dimensions of the lower dimension vector is 512 dimensions. FIG. 24 provides an example dimension reduction process **2401** that may be used to implement operation **2202**.

(124) The process **2201** may include an operation **2203** of subtracting the component of the frame vector in the direction of a vector of ones. The component of the resulting vector from operation **2202** in the direction of the vector of ones is subtracted from itself to produce a new vector that is orthogonal to the vector of ones.

(125) The process **2201** may include an operation **2204** of converting the format of the vector output by operation **2203**. In an example implementation, the operation **2204** may include converting a 64-bit floating point representation of the vector output by operation **2203** to a 16-bit or 32-bit floating-point precision representation. The representations may be converted to a different type of value for storage and computational purposes. The value obtained by converting the representation may be provided as the output **2204** of the process **2201** as the fine-grain fingerprint.

(126) FIG. **23** is a diagram of a process **2301** for coarse-grained fingerprint generation. The process **2301** may be implemented as an alternative to the coarse-grain fingerprinting pipeline **501** of the fingerprint generation process shown in FIG. **5**.

(127) The process **2301** may include an operation **2302** of combining dimensions to produce a lower dimension vector. In one implementation, the number of dimensions of the lower dimension vector is 64 dimensions. FIG. **24** provides an example dimension reduction process **2401** that may be used to implement operation **2302**.

(128) The process **2301** may include an operation **2303** of subtracting the component of the vector from operation **2302** in the direction of a vector of ones. The component of the resulting vector from operation **2302** in the direction of the vector of ones is subtracted from itself to produce a new normalized vector that is orthogonal to the vector of ones.

(129) The process **2301** may include an operation **2304** of matrix multiplying the normalized vector from operation **2303** and a matrix of hyperplanes resulting in a new vector of length P , where P is the number of hyperplanes. The orientation of the hyperplanes can either be precomputed randomly, selected, or discovered using machine learning methods such as gradient descent. The hyperplanes matrix is a predetermined set of unit vectors. The matrix multiplication results in the vector P of cosine distances. Some implementations may use random orientations of unit vectors orthogonal to the ones vector. As a result, the operation **2304** may function similar to a random projections strategy for dimensionality reduction.

(130) The process **2301** may include an operation **2305** of converting the vector from operation **2301** into a bit vector by thresholding. The absolute value of each element of the vector may be compared to a threshold value. If the absolute value is greater than the threshold, then the bit value may be set to 1. Otherwise, the bit value is assigned to 0. The resulting bit value may be provided as output **2307** by the process **2301**. The output **2307** is the coarse-grain fingerprint.

(131) FIG. **24** is an example dimension reduction process **2401**. The process **2401** is performed on the input **2404** which is a vector of length L . The vector is reshaped into a $M \times N$ matrix in operation **2402**, where M and N are positive integers and M represents the number of rows and N represents the number of columns of the matrix. If the desired length of the new vector is not a divisor of L , the next largest divisor of L may be used and any additional elements in the resulting vector may be truncated. The dimensionally reduced vector **2405** may be determined by summing the vector values along the first axis, which results in the vector **2405** of length M . The vector **2405** is provided as output of the process **2401**.

(132) FIG. **25** is a process **2501** for generating a color-sensitive fingerprint. The process **2501** may be used to detect minute differences in coloration between the proxy footage and source footage that may not otherwise be detectable by human users.

(133) The process **2501** may include an operation **2502** of resizing the image frame **2508** received as input. The image frame **2508** may be resized to a standardized size, such as but not limited to 256×256 pixels. Other implementations may resize the image frame **2508** to a different standard size.

(134) The process **2501** may include an operation **2503** in which weighting is applied to each channel of the color space of the resized image. The pixel values of each color channel may be multiplied by a predetermined weight. In one implementation, the image implements the Red Green Blue (RGB) color space and multiplies each channel by the predetermined weight [0.1, 0.7,

0.2]. Each channel is multiplied by a particular weight to scale the pixel value associated with that channel. The example weights and color space shown here show one possible implementation, but other color spaces and channel weighting may be applied in other implementations.

(135) The process **2501** may include an operation **2504** of flattening the matrix representing the resized and weighted image output by operation **2503** into a vector.

(136) The process **2501** may include an operation **2505** of shuffling the elements of the vector produced by the operation **2504**. The elements of the vector are shuffled using a predetermined shuffling routine so that the same reordering occurs each time that the operation **2505** is performed on a vector. The shuffling of the elements ensures that any correlation between pixel adjacency and vector element adjacency is eliminated.

(137) The process **2501** may include an operation **2506** of reducing the dimensionality of the vector. FIG. **24** provides an example dimension reduction process **2401** that may be used to implement operation **2302**.

(138) The process **2501** may include an operation **2507** of folding the vector output by operation **2506**. An example of the folding process is shown in FIG. **12**. The folding process may fine tune the sensitivity of the fingerprint matching and improve the storage requirements for the fingerprint. The fingerprint **2509** may be output by the process **2501**.

(139) FIG. **26** is a diagram of a process **2601** for generating a visual representation of a fingerprint. The process **2601** may take a fine-grain fingerprint as the input **2604**. The process **2601** may include an operation **2602** of combining dimensions of the fingerprint to reduce the number of dimensions of the fingerprint to be represented in the visualization of the fingerprint. The process **2601** may include an operation **2603** of generating a visual graphical representation of the fingerprint. The visual representation may be a 2D or 3D representation of the fingerprint that a user may view to understand attributes of the fingerprint. FIGS. **19A**, **19B**, **19C**, and **19D** show three additional types of visualizations of fingerprints that may be provided. FIGS. **19A**, **19B**, and **19C** are 3D examples of fingerprints. FIG. **19D** is an example of a 2D visualization **1901** that shows a plot **1902** of the attributes of a references frame from the proxy footage and a plot **1903** of a candidate frame from the source footage. The visualization **1901** provides a visual representation of the attributes of the two frames that allows a user to quickly identify similarities and differences between the two frames of footage. While the fingerprint example shown in FIG. **19D** shows a comparison of a frame of proxy footage with a frame of source footage, the fingerprint techniques and the visualizations generated therefrom may be used to provide a visual comparison of the attributes of other types of images as well.

(140) FIG. **27** is a diagram of a process **2701** for resizing and cropping a frame of footage. The process **2701** may receive an input frame **2702** of any resolution. The process **2701** includes two operations that may be performed on the input frame **2702** in any order. The operation **2703** is a first cropping of the input frame **2702**. In the first operation, the input frame is cropped in a tall aspect ratio. The specific aspect ratio represents the minimum aspect ratio for which matching is supported. A common number may be a 4:3 aspect ratio. Next the cropped image is resized to be as small as possible such that the number of pixels is still greater than a constant A such as A=250,000. The operation **2704** is a second cropping of the input frame **2701**. In the second operation **2704**, the input frame is cropped and scaled as in the first operation except using an ultra-wide aspect ratio such as 3:1. This aspect ratio represents the widest possible aspect ratio with which AFCM system may perform matches. The two resized and cropped images may be used in the various fingerprinting and matching techniques provided in the preceding examples.

(141) FIG. **28** is a flow chart of an example process **2800** for matching proxy footage **120** to source footage **117**. The process **2800** may be implemented by the AFCM system **100** discussed in the preceding examples.

(142) The process **2800** may include an operation **2805** of obtaining first video footage that includes a plurality of first video frames of a first video clip. The first video footage may be proxy

footage **120** as discussed in the preceding examples or other footage for which matching footage is being sought. The first video footage may include a plurality of frames of video and at least one video clip for which matching footage is being sought. The first video footage may be video footage that has been selected by a user and for which another version of the video footage is being sought. For example, the user may be attempting to identify source footage from which the first video footage has been derived.

(143) The process **2800** may include an operation **2810** of obtaining second video footage that includes a plurality of second video frames. The second video footage may be source footage **120** as discussed in the preceding examples. The second video footage may include a plurality of frames. The second video footage may include frames associated with a plurality of scenes captured for a video production. The second video footage may be higher-resolution footage than the first video footage, and the first video footage may be derived from the second video footage.

(144) The process **2800** may include an operation **2815** of analyzing the first video footage to generate a plurality of first fingerprints representing each frame of the plurality of first video frames. The fingerprint generation unit **2901** shown in FIG. **29** may be configured to analyze each frame of the first video footage and to generate a fingerprint for each of the frames according to the fingerprint generation techniques provided herein. The fingerprint generation unit **2901** may store the plurality of first fingerprints in the fingerprint indexed datastore **304**.

(145) The process **2800** may include an operation **2820** of analyzing the second video footage to generate a plurality of second fingerprints representing each frame of the plurality of second video frames. The fingerprint generation unit **2901** may also generate fingerprints for each of the frames of the second video footage. The fingerprint generation unit **2901** may also store the plurality of first fingerprints in the fingerprint indexed datastore **304**.

(146) The process **2800** may include an operation **2825** of comparing the first fingerprints to the second fingerprints to identify one or more candidate clip matches for the first video clip in the second video footage. The matching unit **2902** shown in FIG. **29** may perform a matching operation to identify one or more candidates for the first video clip as discussed in the examples described herein.

(147) The process **2800** may include an operation **2830** of comparing the first fingerprints to the second fingerprints to generate similarity scores representing how similar each of the plurality of first video frames are to the plurality of second video frames. The matching unit **2902** may calculate a score matrix and/or one or more pseudo-probability matrices that represent how similar each of the first video frame and the second video frames are. The similarity scores and/or the one or more pseudo-probability matrices may be determined using the various matching techniques described herein.

(148) The process **2800** may include an operation **2835** of identifying one or more candidate clip matches for the first video clip based on the similarity scores. Once the similarity scores have been determined, the matching unit **2902** may identify one or more clips of video footage that may be a match for the first video clip. The matching unit **2902** may identify more than one potential match for the first video clip, because the second video footage may include multiple clips that may be quite similar. For example, a scene in a video production may be captured in multiple takes which are quite similar, and the first video clip may be derived from one of these takes.

(149) The process **2800** may include an operation **2840** of selecting a best match for the first video clip in the second video footage from the one or more candidate clip matches for the first video clip. The matching unit **2902** may select a best match for the first video clip based on the various techniques discussed herein. For example, where a scene has been captured in multiple takes and the first video clip is derived from footage of one of these takes, the matching unit **2902** may identify the subtle similarities between the first video clip and a particular take to identify the correct source footage.

(150) The process **2800** may include, in the operation **2840**, presenting the best match for the first

video clip on a user interface of the data processing system. The AFCM system **100** may provide a user interface for displaying the video clip of the second video footage which matched the first video clip. The user interface may provide means for the user to saving a copy of the video clip of the second video clip, to generate a report identifying a location of the first video clip in the second video footage, and/or to perform other actions on the video clip of the second video footage.

(151) FIG. **29** is a diagram showing details of fingerprint generation and using the fingerprints generated to perform a matching operation. The matching operation in this example is performed on first media **2904** and second media **2905**. The first media **2904** may be first video footage that includes one or more frames of footage for which a match in the second media **2905** is being sought. The second media **2905** may be second video footage that includes multiple frames of footage. The first media **2904** may include frames associated with one or more clips of video footage that are being sought in the footage of the second media **2905**. The first media **2904** may be proxy footage **117** or other footage for which a matching frames and/or clips of footage in the second media **2905** are being sought. The second media **2905** may be source footage **117** and/or other footage in which frames and/or clips of footage matching those of the first media **2904** are being sought. The AFCM system **100** may provide a user interface configured to allow a user to specify the first media **2904** identifying the content being sought and the second media **2905** to be searched for matches.

(152) In the example shown in FIG. **29**, the first media **2904** and second media **2905** are ingested by the AFCM system **100** and provided to the fingerprint generation unit **2901**. The fingerprint generation unit **2901** may implement the process **500** shown in FIG. **5** for generating a fingerprint of a source image, such as a frame of proxy footage **120** or source footage **117**. The fingerprint generation unit **2901** may be configured to generate a fingerprint for each of the frames of footage, and to store the fingerprints in the fingerprint indexed datastore **304**.

(153) In this example, the fingerprints **2908** for the first media **2904** may be passed to the matching unit **2902** and the match unit **2902** may search the fingerprint indexed datastore **304** for matches from the second media **2905**. The first media **2904** may be proxy footage **117** and/or other footage for which matching frames and/or clips are being sought in the second media **2904**. The matching unit **2902** may be configured to identify one or more frames, and/or one or more clips of the second media **2905** that match the first media **2904**. The matching unit **2902** may generate a match list and reports **2907**. The match list identifies the locations of the matching frames and/or clips of the second media **2905** that match one or more frames or clips of the first media **2904**. The match unit **2902** may also generate reports identifying frames and/or clips included in the first media **2904** for which a match could not be found in the second media **2905**. Additional details of the matching operations that may be performed by the matching unit **2902** are provided in the examples which follow.

(154) FIG. **30** is a diagram that shows additional details of the matching operations that may be performed by the matching unit **2902**. In this example, the first media **2904** is proxy footage which has been converted to fingerprints **3004** by the fingerprint generation unit **2901**, and the second media **2905** is source footage which has been converted to fingerprints **3005** by the fingerprint generation unit **2901**. The fingerprints **3004** and **3005** have been stored in the fingerprint indexed datastore **304**. The fingerprints **3004** include p fingerprints, where p is a positive integer. The fingerprints **3004** include a first proxy fingerprint **3002** representing the first frame of the proxy footage and a final proxy fingerprint **3011** representing the final frame of the proxy footage. The fingerprints **3005** include y fingerprints, where y is a positive integer. The fingerprints **3005** include a first source fingerprint **3003** representing the first frame of the source footage and a final source fingerprint **3010** representing the final frame of the proxy footage. The matching unit **2902** analyzes the fingerprints **3004** representing the proxy footage and attempts to find matches in the fingerprints **3005** representing the source footage. The matching unit **2902** may take into account threshold settings **3007** for determining how similar a fingerprint of the proxy footage and the

source footage must be before determining that the frame of proxy footage is a match candidate. Additional details of the threshold settings **3007** are discussed with respect to the example implementations which follow. The matching unit **2902** may output the match list and reports **2907** as discussed in the preceding examples.

(155) FIG. **31** is a diagram of a process **3101** for performing a matching operation. The matching unit **2902** may be configured to perform the process **3101** shown in FIG. **31** to identify matches between one or more clips and/or frames of proxy footage **120** and source footage **120** in which frames and/or clips included in the proxy footage **120** are being sought.

(156) The process **3101** receives an input **3107** that identifies the proxy footage **120** and the source footage **117** for which the matching operation is to be performed. As discussed in the preceding examples, the AFCM system **100** may provide a user interface that is configured to allow the user to identify the proxy footage **120** and the source footage **117** for which the matching operation is to be performed. The proxy footage **120** and the source **117** are processed by the fingerprint generation unit **2901** to generate the fingerprints of the proxy footage **120** and the source footage **117** if the fingerprints have not already been generated for the footage. The fingerprints may be stored in the fingerprint indexed datastore **304**. The matching unit **2902** may be configured to obtain the fingerprints from the fingerprint indexed datastore **304** and provide them as an input to the process **3101**.

(157) The process **3101** may include an operation of calculating a score matrix **3102**. A score matrix **Sis** an $M \times N$ matrix, where M represents a number of frames included in the proxy footage **120** and N represents a number of frames included in the source footage **117**. Each cell $S_{sub.ij}$ of the score matrix includes a similarity score of the i th proxy frame with the j th source frame. The similarity score for each cell is determined based on the fingerprints associated with the proxy frame and source frame associated with that cell of the score matrix. The similarity score for each cell may be determined based on the cosine distance between the proxy frame and the source frame associated with the cell.

(158) FIG. **34** is a diagram of an example score matrix **3403**. The rows of the score matrix **3403** each represent a proxy frame **3402** of the proxy footage **117**. The columns of the score matrix **3403** each represent a respective source frame **3401** of the source footage **120**. Each proxy frame **3402** is compared to each source frame **3401**, and a similarity score is included populated in cells where the proxy frame and the source frame exceed a similarity threshold. Example score **3404** represents the similarity score of proxy frame **4** and source frame **8**. The cells which are blank are associated with a proxy frame and source frame pair that was not sufficiently similar to cause the matching unit **2902** to determine a similarity score. As discussed above, an LSH operation or other heuristic means may be used to determine whether the source frame and the proxy frame are sufficiently similar to cause the matching unit **2902** to determine the similarity score. In other implementations, a similarity score may be determined for each of the cells of the score matrix **3403**.

(159) In some implementations, the score matrix may be a sparse matrix, and a similarity score may only be calculated for the fingerprints of the proxy frame and the source frame pairs that are sufficiently similar. This determination may be performed heuristically. In some implementations, an LSH operation may be applied to the fingerprints of the proxy frame and the source frame to determine whether the two frames are sufficiently similar for determining a similarity score. The LSH operation may group similar frames into a same “bucket” and group frames that are not similar into separate buckets. If the proxy frame and the source frame for a particular cell are grouped into the same bucket, the similarity score for that cell of the score matrix may be determined. Other threshold calculations may be performed for determining whether the proxy and source frames are sufficiently similar for determining the similarity score.

(160) FIG. **32** is a diagram of an example process **3201** that may be used to calculate the score matrix in operation **3102** of process **3101**. The input **3207** to the process **3201** includes the fingerprints of the proxy footage **120** and the source footage **117**. The process **3201** may perform

an operation of finding initial candidates **3202** in which at least a subset of the pairs of proxy frames and source frames that may be included in the score matrix is selected based on the respective proxy frame and source frame satisfying a similarity threshold. As discussed above, the initial candidates may be selected by performing an LSH operation on the proxy frames and source frames to identify pairs of source and proxy frames to serve as candidates for further processing. The process **3201** may also include an operation of calculating a similarity score for the pairs of source and proxy frames selected in operation **3203**. The output **3208** of the process **3201** is a score matrix, such as the score matrix **3403** shown in FIG. **34**.

(161) Returning now to FIG. **31**, the process **3101** may include an operation of calculating one or more pseudo-probability matrices **3103**. The purpose of the pseudo-probability matrixes is to represent the probability that any given frame level match candidate is indeed a match. Multiple pseudo-probability matrices may be calculated to account for potential inaccuracies in one or more of the calculated pseudo-probability matrices. In some implementations, the multiple pseudo-probability matrices may include a normalized pseudo-probability matrix and a denormalized pseudo-probability matrix. The pseudo-probability matrices may be calculated using one or more algorithms for estimating the probability that a particular frame level candidate match is a match. The particular algorithm used for determining the probability may vary from implementation to implementation. A normalized and denormalized pseudo-probability matrix may be calculated for each of the algorithms used to estimate the probability for each frame level candidate match. The estimated probabilities output by a respective algorithm may be used to generate the denormalized pseudo-probability matrix, and the normalized pseudo-probability matrix may be derived from the denormalized pseudo-probability matrix. In the normalized pseudo-probability matrix, the values of each of the cells of each column may be modified so that the sum of the values of each column is equal to 100%.

(162) FIG. **35** is an example of denormalized probability matrix **3503**. The rows of the probability matrix **3503** each represent a proxy frame **3502** of the proxy footage **117**. The columns of the score matrix **3503** each represent a respective source frame **3501** of the source footage **120**. The probability values included in the cells of the probability matrix **3503**, such as the example entry **3504**, represent a probability that the proxy frame and source frame associated with that cell are a match. A corresponding normalized probability matrix may be generated from the denormalized probability matrix by adjusting the values of the denormalized probability matrix so that each column of values adds up to 100%.

(163) Returning again to FIG. **31**, the process **3101** may include an operation **3104** of ordering the proxy frames by descending pseudo-probability score. For each row of the probability matrix, such as the denormalized probability matrix **3503**, the values may be sorted from highest to lowest probability matches with a respective one of the source frames. The maximum probability value may be selected for each row representing a proxy frame, and then the remaining pairs of proxy frames and source frames may be sorted by probability. For example, referring to the denormalized probability matrix **3503**, the maximum value for each frame is represented by a x, y, and p triplet, where x represents the proxy frame row, y represents the source frame column, and p represents the probability value. Thus, the values for the denormalized probability matrix **3503** would be: [0, 0, 50%], [1, 1, 80%], [2, 2, 90%], [3, 5, 70%], and [4, 8, 80%]. The values may then be reordered according to the probability of a match to be: [4, 8, 80%], [3, 5, 70%], [2, 2, 90%], [1, 1, 80%], and [0,0, 50%].

(164) In instances where both a normalized and denormalized probability matrix are determined, the values each cell of the two matrices may be averaged to determine an overall probability score. In some instances, a weighted average may be used in which either the normalized or denormalized values may be weighted more heavily. Furthermore, in instances where there are more than one normalized and denormalized probability matrices calculated based on the results of multiple algorithms being used to calculate the probabilities, the values of each cell of the multiple

denormalized matrices may be averaged to determine an overall denormalized probability matrix, and values of each cell of the multiple normalized probability matrices may be averaged to determine an overall normalized probability matrix. The overall denormalized and normalized may be averages to determine an overall score for each cell. Other techniques for determining a score for each pair of proxy and source frames may be used in other implementations.

(165) The process **3101** may include an operation of finding the best clip candidates **3105**. The proxy frame may be part of a clip that includes more than one frame. The clips may be identified using the processes shown in FIG. **33**. The process **3101** may then provide the clip candidates selected by operation **3105** as output **3108**. The clip candidates may be presented on a user interface of the AFCM system **100** and/or included in a match report generated by the matching unit **2902**.

(166) FIG. **33** is a diagram of a process **3301** that may be used to implement operation **3105** for finding a best clip candidate in the process **3101** shown in FIG. **31**. The process **3301** may be implemented by the matching unit **2902**. The process **3301** may receive an input **3307** which may include the proxy frames ordered by descending pseudo-probability score output by operation **3104** of the process **3101**.

(167) The process **3301** may include an operation **3302** of filtering frame level matches. At least a portion of the candidate frames may be discarded based on the similarity score, the probability matrix or matrices values associated with the frame, or both. The matching unit **2902** may determine whether to discard a candidate from based on threshold information stored in the threshold settings **3007**. The threshold settings **3007** may be stored in a persistent memory of the AFCM system **100** and may be configurable by users of the AFCM system **100**. The AFCM system **100** may provide a user interface that permits a user to configure the threshold settings to be used for determining whether to discard potential matches. The user interface may provide guidance to the user for selecting an appropriate threshold value or set of values for the needs of the user.

(168) The process **3301** may include an operation **3303** of calculating a clip from a point match of a frame of proxy footage. The clip may be identified by searching for other matching frames that are adjacent to a matching frame. FIGS. **36-39** demonstrate how a clip associated with a matching frame may be identified. The operation **3303** may potentially identify multiple matching clips.

(169) The process **3301** may include an operation of calculating clip scores **3304**. Each clip candidate may be analyzed and assigned a clip score that represents how well the clip of proxy footage **120** matches that of the source footage **117**. Multiple clips may be very similar, and the scoring process may be used to identify which clip of the source footage **117** is most similar to that of the proxy footage **120**. In some implementations, the clip score for the frame may be determined by summing the frame-level similarity scores for each of the frames of the clip. Other techniques for calculating the frame clip scores may also be used in addition to or instead of summing of the frame-level similarity scores.

(170) The process **3301** may include an operation of selecting the clip with the best clip score **3305** and outputting the selected clip as output **3308**. In some implementations, the clip having a clip score that represent a best match between the clip of proxy footage **120** and the clip of source footage **117** may be selected. This approach may be used to identify the clip of the source footage **117** that provides the closest match for that of the clip of the proxy footage **120**. The differences between clip candidates may be very subtle and would be difficult and time consuming for a human user to attempt to identify the best match for the proxy footage using current tools. The approach provided in the process **3301** may quickly and accurately find the best match.

(171) FIGS. **36-39** are diagrams showing examples of processes for identifying clips associated with a frame. As indicated above, a frame of proxy footage **120** that matches to a frame of the source footage **117** may be part of a clip that includes multiple claims of footage. Where the frame rate of the proxy footage **120** and the frame rate of the source footage **117** have a 1:1 correspondence, the clips may be identified by searching along diagonals within the score matrix

3503 and/or the pseudo-probability matrices to identify other frames of the proxy footage **120** and source footage **117** that matched. The clips may be identified by analyzing values along the diagonals of the matrices as shown in the examples which follow. Adjacent matching frames will be disposed along the diagonals of the matrices where the frame rates of the proxy footage **120** and the source footage **117** are the same.

(172) FIG. **36** shows an example of how the matrix values along a diagonal may be analyzed to identify clip candidates from the source footage **117** that match the clips in the proxy footage **120**. In the example shown in FIG. **36**, an array **3601** of values from a diagonal set of cells of a score matrix, such as the score matrix **3503**. The length of the array **3601** depends upon the size of the score matrix, which in turn depends upon the number of proxy frames **120** and the number of source frames **117** that have been compared to generate the score matrix. The diagonal may be selected from the score matrix where the diagonal includes one or more non-blank cells that include a similarity score determined as discussed in the preceding examples. In the example shown in FIG. **36** there are similarity scores, such as the example score **3608**, for each of the elements of the array. However, the score matrix may be sparsely populated, and at least some of the cells may be blank if the similarity of the proxy frame and source frame associated with that cell did not meet the threshold conditions discussed in the preceding examples. In the example shown in FIG. **36**, the similarity score is lower for cells where the proxy frame and the source frame are more similar.

(173) The values of the array **3601** may be evaluated to identify the frames comprising the clip. The similarity values at cell preceding cell **3603** and the similarity score of the cell following cell **3604** show a distinct change in the similarity scores. The similarity scores of the cell preceding cell **3603** and the similarity score of the cell following cell **3604** are significantly higher than the similarity scores of the cells **3603** and **3604**. Furthermore, there are no substantial changes in the similarity scores of the cells between the cells **3603** and **3604**. Therefore, the frames associated with the cell **3603** through and including cell **3604** form a single matching candidate clip **3602**. The difference between the similarity scores associated with non-matching frames can be represented by the higher values **3606** and the lower values **3607** represent the matching claims which make up the candidate clip.

(174) FIGS. **37-39** show examples in which the candidate clips are marked in a matrix of the same size as the score matrix. The frames identified by the process shown in FIG. **36** may be used to mark the cells corresponding to frames that make up a candidate clip in which source and proxy frames matched. In FIGS. **37-39**, the frames that are shaded black are the frames for which a match between the proxy footage **120** and the source footage **117** were found.

(175) FIG. **37** shows an example matrix **3701** in which the matches for the proxy frames **3702** and the source frames **3703** have been shaded. The matrix **3701** may be a sparse matrix in which the shaded cells include a value indicating a match between the respective proxy frame and source frame associated with that cell. The cell **3705** has been left unshaded because the matching unit **2902** determined that the proxy frame and the source frame pair [4, 2] were not a match. However, the cell **3704** has been shaded because the matching unit **2902** determined the proxy frame and the source frame pair [4, 6] were a match. In some implementations, the matching cells may be set with a 1-bit value indicating there is a match. The shading shown in FIG. **37** is merely intended to demonstrate the matching process described herein. The example matrices shown in FIGS. **37-39** may be utilized internally by the matching unit **2902** for performing the matching operations described herein, and the AFCM **100** may not present a representation of the matrices on a user interface of the AFCM **100**.

(176) In the example shown in FIG. **37**, a single candidate clip that includes five frames has been identified by the matching operation, which includes the frames associated with cells [0, 2], [1, 3], [2, 4], [3, 5], and [4, 6]. The length of a matched clip may include a different number of frames. The specific number of frames matched in this example and the examples shown in FIGS. **38** and

39 demonstrate examples of the matching techniques provided herein and do not limit the matching techniques to these specific matches.

(177) FIG. **38** shows an example matrix **3801** in which the matches for the proxy frames **3802** and the source frames **3803** have been shaded. Two clips **3804** and **3806** have been identified in this example. The clip **3804** includes two frames and the clip **3806** includes three frames. As previously indicated, the length of the clips matched may vary and are not limited to these specific examples. For example, a clip may include 24 or more frames per second, and thus, the number of frames of a matched clip may be much longer than the examples shown in FIGS. **37-38**, depending upon the frame rate of the footage that is being analyzed for a match.

(178) FIG. **39** shows an example matrix **3901** in which the matches for the proxy frames **3902** and the source frames **3903** have been shaded. Two clips **3904** and **3905** have been identified in this example. The example shown in FIG. **39** includes a row **3907** for where there was no match found for the proxy frame **2**. The proxy frame **2** may be identified by the matching unit **2902** and included in the match list and reports **2907**.

(179) The detailed examples of systems, devices, and techniques described in connection with FIGS. **1-39** are presented herein for illustration of the disclosure and its benefits. Such examples of use should not be construed to be limitations on the logical process embodiments of the disclosure, nor should variations of user interface methods from those described herein be considered outside the scope of the present disclosure. It is understood that references to displaying or presenting an item (such as, but not limited to, presenting an image on a display device, presenting audio via one or more loudspeakers, and/or vibrating a device) include issuing instructions, commands, and/or signals causing, or reasonably expected to cause, a device or system to display or present the item. In some embodiments, various features described in FIGS. **1-39** are implemented in respective modules, which may also be referred to as, and/or include, logic, components, units, and/or mechanisms. Modules may constitute either software modules (for example, code embodied on a machine-readable medium) or hardware modules.

(180) In some examples, a hardware module may be implemented mechanically, electronically, or with any suitable combination thereof. For example, a hardware module may include dedicated circuitry or logic that is configured to perform certain operations. For example, a hardware module may include a special-purpose processor, such as a field-programmable gate array (FPGA) or an Application Specific Integrated Circuit (ASIC). A hardware module may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations and may include a portion of machine-readable medium data and/or instructions for such configuration. For example, a hardware module may include software encompassed within a programmable processor configured to execute a set of software instructions. It will be appreciated that the decision to implement a hardware module mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (for example, configured by software) may be driven by cost, time, support, and engineering considerations.

(181) Accordingly, the phrase “hardware module” should be understood to encompass a tangible entity capable of performing certain operations and may be configured or arranged in a certain physical manner, be that an entity that is physically constructed, permanently configured (for example, hardwired), and/or temporarily configured (for example, programmed) to operate in a certain manner or to perform certain operations described herein. As used herein, “hardware-implemented module” refers to a hardware module. Considering examples in which hardware modules are temporarily configured (for example, programmed), each of the hardware modules need not be configured or instantiated at any one instance in time. For example, where a hardware module includes a programmable processor configured by software to become a special-purpose processor, the programmable processor may be configured as respectively different special-purpose processors (for example, including different hardware modules) at different times. Software may accordingly configure a processor or processors, for example, to constitute a particular hardware

module at one instance of time and to constitute a different hardware module at a different instance of time. A hardware module implemented using one or more processors may be referred to as being “processor implemented” or “computer implemented.”

(182) Hardware modules can provide information to, and receive information from, other hardware modules. Accordingly, the described hardware modules may be regarded as being communicatively coupled. Where multiple hardware modules exist contemporaneously, communications may be achieved through signal transmission (for example, over appropriate circuits and buses) between or among two or more of the hardware modules. In embodiments in which multiple hardware modules are configured or instantiated at different times, communications between such hardware modules may be achieved, for example, through the storage and retrieval of information in memory devices to which the multiple hardware modules have access. For example, one hardware module may perform an operation and store the output in a memory device, and another hardware module may then access the memory device to retrieve and process the stored output.

(183) In some examples, at least some of the operations of a method may be performed by one or more processors or processor-implemented modules. Moreover, the one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by, and/or among, multiple computers (as examples of machines including processors), with these operations being accessible via a network (for example, the Internet) and/or via one or more software interfaces (for example, an application program interface (API)). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across several machines. Processors or processor-implemented modules may be in a single geographic location (for example, within a home or office environment, or a server farm), or may be distributed across multiple geographic locations.

(184) FIG. **40** is a block diagram **4000** illustrating an example software architecture **4002**, various portions of which may be used in conjunction with various hardware architectures herein described, which may implement any of the above-described features. FIG. **40** is a non-limiting example of a software architecture, and it will be appreciated that many other architectures may be implemented to facilitate the functionality described herein. The software architecture **4002** may execute on hardware such as a machine **4100** of FIG. **41** that includes, among other things, processors **4110**, memory **4130**, and input/output (I/O) components **4150**. A representative hardware layer **4004** is illustrated and can represent, for example, the machine **4100** of FIG. **41**. The representative hardware layer **4004** includes a processing unit **4006** and associated executable instructions **4008**. The executable instructions **4008** represent executable instructions of the software architecture **4002**, including implementation of the methods, modules and so forth described herein. The hardware layer **4004** also includes a memory/storage **4010**, which also includes the executable instructions **4008** and accompanying data. The hardware layer **4004** may also include other hardware modules **4012**. Instructions **4008** held by processing unit **4006** may be portions of instructions **4008** held by the memory/storage **4010**.

(185) The example software architecture **4002** may be conceptualized as layers, each providing various functionality. For example, the software architecture **4002** may include layers and components such as an operating system (OS) **4014**, libraries **4016**, frameworks **4018**, applications **4020**, and a presentation layer **4044**. Operationally, the applications **4020** and/or other components within the layers may invoke API calls **4024** to other layers and receive corresponding results **4026**. The layers illustrated are representative in nature and other software architectures may include additional or different layers. For example, some mobile or special purpose operating systems may not provide the frameworks/middleware **4018**.

(186) The OS **4014** may manage hardware resources and provide common services. The OS **4014** may include, for example, a kernel **4028**, services **4030**, and drivers **4032**. The kernel **4028** may act as an abstraction layer between the hardware layer **4004** and other software layers. For example,

the kernel **4028** may be responsible for memory management, processor management (for example, scheduling), component management, networking, security settings, and so on. The services **4030** may provide other common services for the other software layers. The drivers **4032** may be responsible for controlling or interfacing with the underlying hardware layer **4004**. For instance, the drivers **4032** may include display drivers, camera drivers, memory/storage drivers, peripheral device drivers (for example, via Universal Serial Bus (USB)), network and/or wireless communication drivers, audio drivers, and so forth depending on the hardware and/or software configuration.

(187) The libraries **4016** may provide a common infrastructure that may be used by the applications **4020** and/or other components and/or layers. The libraries **4016** typically provide functionality for use by other software modules to perform tasks, rather than interacting directly with the OS **4014**. The libraries **4016** may include system libraries **4034** (for example, C standard library) that may provide functions such as memory allocation, string manipulation, file operations. In addition, the libraries **4016** may include API libraries **4036** such as media libraries (for example, supporting presentation and manipulation of image, sound, and/or video data formats), graphics libraries (for example, an OpenGL library for rendering 2D and 3D graphics on a display), database libraries (for example, SQLite or other relational database functions), and web libraries (for example, WebKit that may provide web browsing functionality). The libraries **4016** may also include a wide variety of other libraries **4038** to provide many functions for applications **4020** and other software modules.

(188) The frameworks **4018** (also sometimes referred to as middleware) provide a higher-level common infrastructure that may be used by the applications **4020** and/or other software modules. For example, the frameworks **4018** may provide various graphic user interface (GUI) functions, high-level resource management, or high-level location services. The frameworks **4018** may provide a broad spectrum of other APIs for applications **4020** and/or other software modules.

(189) The applications **4020** include built-in applications **4040** and/or third-party applications **4042**. Examples of built-in applications **4040** may include, but are not limited to, a contacts application, a browser application, a location application, a media application, a messaging application, and/or a game application. Third-party applications **4042** may include any applications developed by an entity other than the vendor of the particular platform. The applications **4020** may use functions available via OS **4014**, libraries **4016**, frameworks **4018**, and presentation layer **4044** to create user interfaces to interact with users.

(190) Some software architectures use virtual machines, as illustrated by a virtual machine **4048**. The virtual machine **4048** provides an execution environment where applications/modules can execute as if they were executing on a hardware machine (such as the machine **4100** of FIG. **41**, for example). The virtual machine **4048** may be hosted by a host OS (for example, OS **4014**) or hypervisor, and may have a virtual machine monitor **4046** which manages operation of the virtual machine **4048** and interoperation with the host operating system. A software architecture, which may be different from software architecture **4002** outside of the virtual machine, executes within the virtual machine **4048** such as an OS **4050**, libraries **4052**, frameworks **4054**, applications **4056**, and/or a presentation layer **4058**.

(191) FIG. **41** is a block diagram illustrating components of an example machine **4100** configured to read instructions from a machine-readable medium (for example, a machine-readable medium) and perform any of the features described herein. The example machine **4100** is in a form of a computer system, within which instructions **4116** (for example, in the form of software components) for causing the machine **4100** to perform any of the features described herein may be executed. As such, the instructions **4116** may be used to implement modules or components described herein. The instructions **4116** cause unprogrammed and/or unconfigured machine **4100** to operate as a particular machine configured to carry out the described features. The machine **4100** may be configured to operate as a standalone device or may be coupled (for example, networked)

to other machines. In a networked deployment, the machine **4100** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a node in a peer-to-peer or distributed network environment. Machine **4100** may be embodied as, for example, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a gaming and/or entertainment system, a smart phone, a mobile device, a wearable device (for example, a smart watch), and an Internet of Things (IoT) device. Further, although only a single machine **4100** is illustrated, the term “machine” includes a collection of machines that individually or jointly execute the instructions **4116**.

(192) The machine **4100** may include processors **4110**, memory **4130**, and I/O components **4150**, which may be communicatively coupled via, for example, a bus **4102**. The bus **4102** may include multiple buses coupling various elements of machine **4100** via various bus technologies and protocols. In an example, the processors **4110** (including, for example, a central processing unit (CPU), a graphics processing unit (GPU), a digital signal processor (DSP), an ASIC, or a suitable combination thereof) may include one or more processors **4112a** to **4112n** that may execute the instructions **4116** and process data. In some examples, one or more processors **4110** may execute instructions provided or identified by one or more other processors **4110**. The term “processor” includes a multi-core processor including cores that may execute instructions contemporaneously. Although FIG. **41** shows multiple processors, the machine **4100** may include a single processor with a single core, a single processor with multiple cores (for example, a multi-core processor), multiple processors each with a single core, multiple processors each with multiple cores, or any combination thereof. In some examples, the machine **4100** may include multiple processors distributed among multiple machines.

(193) The memory/storage **4130** may include a main memory **4132**, a static memory **4134**, or other memory, and a storage unit **4136**, both accessible to the processors **4110** such as via the bus **4102**. The storage unit **4136** and memory **4132**, **4134** store instructions **4116** embodying any one or more of the functions described herein. The memory/storage **4130** may also store temporary, intermediate, and/or long-term data for processors **4110**. The instructions **4116** may also reside, completely or partially, within the memory **4132**, **4134**, within the storage unit **4136**, within at least one of the processors **4110** (for example, within a command buffer or cache memory), within memory at least one of I/O components **4150**, or any suitable combination thereof, during execution thereof. Accordingly, the memory **4132**, **4134**, the storage unit **4136**, memory in processors **4110**, and memory in I/O components **4150** are examples of machine-readable media.

(194) As used herein, “machine-readable medium” refers to a device able to temporarily or permanently store instructions and data that cause machine **4100** to operate in a specific fashion, and may include, but is not limited to, random-access memory (RAM), read-only memory (ROM), buffer memory, flash memory, optical storage media, magnetic storage media and devices, cache memory, network-accessible or cloud storage, other types of storage and/or any suitable combination thereof. The term “machine-readable medium” applies to a single medium, or combination of multiple media, used to store instructions (for example, instructions **4116**) for execution by a machine **4100** such that the instructions, when executed by one or more processors **4110** of the machine **4100**, cause the machine **4100** to perform and one or more of the features described herein. Accordingly, a “machine-readable medium” may refer to a single storage device, as well as “cloud-based” storage systems or storage networks that include multiple storage apparatus or devices. The term “machine-readable medium” excludes signals per se.

(195) The I/O components **4150** may include a wide variety of hardware components adapted to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **4150** included in a particular machine will depend on the type and/or function of the machine. For example, mobile devices such as mobile phones may include a touch input device, whereas a headless server or IoT device may not include such a touch input device. The particular examples of I/O components illustrated in FIG. **41** are in

no way limiting, and other types of components may be included in machine **4100**. The grouping of I/O components **4150** are merely for simplifying this discussion, and the grouping is in no way limiting. In various examples, the I/O components **4150** may include user output components **4152** and user input components **4154**. User output components **4152** may include, for example, display components for displaying information (for example, a liquid crystal display (LCD) or a projector), acoustic components (for example, speakers), haptic components (for example, a vibratory motor or force-feedback device), and/or other signal generators. User input components **4154** may include, for example, alphanumeric input components (for example, a keyboard or a touch screen), pointing components (for example, a mouse device, a touchpad, or another pointing instrument), and/or tactile input components (for example, a physical button or a touch screen that provides location and/or force of touches or touch gestures) configured for receiving various user inputs, such as user commands and/or selections.

(196) In some examples, the I/O components **4150** may include biometric components **4156**, motion components **4158**, environmental components **4160**, and/or position components **4162**, among a wide array of other physical sensor components. The biometric components **4156** may include, for example, components to detect body expressions (for example, facial expressions, vocal expressions, hand or body gestures, or eye tracking), measure biosignals (for example, heart rate or brain waves), and identify a person (for example, via voice-, retina-, fingerprint-, and/or facial-based identification). The motion components **4158** may include, for example, acceleration sensors (for example, an accelerometer) and rotation sensors (for example, a gyroscope). The environmental components **4160** may include, for example, illumination sensors, temperature sensors, humidity sensors, pressure sensors (for example, a barometer), acoustic sensors (for example, a microphone used to detect ambient noise), proximity sensors (for example, infrared sensing of nearby objects), and/or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment. The position components **4162** may include, for example, location sensors (for example, a Global Position System (GPS) receiver), altitude sensors (for example, an air pressure sensor from which altitude may be derived), and/or orientation sensors (for example, magnetometers).

(197) The I/O components **4150** may include communication components **4164**, implementing a wide variety of technologies operable to couple the machine **4100** to network(s) **4170** and/or device(s) **4180** via respective communicative couplings **4172** and **4182**. The communication components **4164** may include one or more network interface components or other suitable devices to interface with the network(s) **4170**. The communication components **4164** may include, for example, components adapted to provide wired communication, wireless communication, cellular communication, Near Field Communication (NFC), Bluetooth communication, Wi-Fi, and/or communication via other modalities. The device(s) **4180** may include other machines or various peripheral devices (for example, coupled via USB). GPU-accelerated computing **4199** may also be provided to facilitate processing-intensive operations, such as but not limited to image analysis, deep learning and/or other machine learning techniques, and/or other computationally intensive tasks.

(198) In some examples, the communication components **4164** may detect identifiers or include components adapted to detect identifiers. For example, the communication components **4164** may include Radio Frequency Identification (RFID) tag readers, NFC detectors, optical sensors (for example, one- or multi-dimensional bar codes, or other optical codes), and/or acoustic detectors (for example, microphones to identify tagged audio signals). In some examples, location information may be determined based on information from the communication components **4162**, such as, but not limited to, geo-location via Internet Protocol (IP) address, location via Wi-Fi, cellular, NFC, Bluetooth, or other wireless station identification and/or signal triangulation.

(199) While various embodiments have been described, the description is intended to be exemplary, rather than limiting, and it is understood that many more embodiments and implementations are

possible that are within the scope of the embodiments. Although many possible combinations of features are shown in the accompanying figures and discussed in this detailed description, many other combinations of the disclosed features are possible. Any feature of any embodiment may be used in combination with or substituted for any other feature or element in any other embodiment unless specifically restricted. Therefore, it will be understood that any of the features shown and/or discussed in the present disclosure may be implemented together in any suitable combination. Accordingly, the embodiments are not to be restricted except in light of the attached claims and their equivalents. Also, various modifications and changes may be made within the scope of the attached claims.

(200) While the foregoing has described what are considered to be the best mode and/or other examples, it is understood that various modifications may be made therein and that the subject matter disclosed herein may be implemented in various forms and examples, and that the teachings may be applied in numerous applications, only some of which have been described herein. It is intended by the following claims to claim any and all applications, modifications and variations that fall within the true scope of the present teachings.

(201) Unless otherwise stated, all measurements, values, ratings, positions, magnitudes, sizes, and other specifications that are set forth in this specification, including in the claims that follow, are approximate, not exact. They are intended to have a reasonable range that is consistent with the functions to which they relate and with what is customary in the art to which they pertain.

(202) The scope of protection is limited solely by the claims that now follow. That scope is intended and should be interpreted to be as broad as is consistent with the ordinary meaning of the language that is used in the claims when interpreted in light of this specification and the prosecution history that follows and to encompass all structural and functional equivalents.

Notwithstanding, none of the claims are intended to embrace subject matter that fails to satisfy the requirement of Sections **101**, **102**, or **103** of the Patent Act, nor should they be interpreted in such a way. Any unintended embracement of such subject matter is hereby disclaimed.

(203) Except as stated immediately above, nothing that has been stated or illustrated is intended or should be interpreted to cause a dedication of any component, step, feature, object, benefit, advantage, or equivalent to the public, regardless of whether it is or is not recited in the claims.

(204) It will be understood that the terms and expressions used herein have the ordinary meaning as is accorded to such terms and expressions with respect to their corresponding respective areas of inquiry and study except where specific meanings have otherwise been set forth herein. Relational terms such as first and second and the like may be used solely to distinguish one entity or action from another without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “a” or “an” does not, without further constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

(205) The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various examples for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claims require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed example. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

Claims

1. A data processing system comprising: a processor; and a machine-readable medium storing executable instructions that, when executed, cause the processor to perform operations comprising: obtaining first video footage that includes a plurality of first video frames of a first video clip; obtaining second video footage that includes a plurality of second video frames; analyzing the first video footage to generate a plurality of first fingerprints representing each frame of the plurality of first video frames, wherein each fingerprint of the plurality of first fingerprints is generated based on a respective video frame of the plurality of first video frames, each fingerprint comprising a first coarse-grain fingerprint element and a first fine-grain fingerprint element, the first coarse-grain fingerprint element being generated from a reduced-sized version of the respective video frame that is smaller than a version of the respective video frame used to generate the first fine-grain fingerprint element, wherein generating the plurality of first fingerprints includes: generating the first coarse-grain fingerprint element based on a frame of the first video footage by: resizing the frame of the first video footage to a first standard image size to generate a first resized frame; subdividing the first resized frame into a plurality of cells; and analyzing each of the plurality of cells using a folding locality-sensitive hashing (LSH) algorithm to generate a plurality of hash values comprising a hash value for each cell of the plurality of cells; and generating the first fine-grain fingerprint element based on a frame of the first video footage by: generating a first vector by reducing a dimensionality of a first vector representation of the frame; generating a second vector orthogonal to a vector of ones by subtracting a component of the first vector in a direction of the vector of ones from the component of the first vector; and converting the second vector into a bit string representing values of elements of the second vector; analyzing the second video footage to generate a plurality of second fingerprints representing each frame of the plurality of second video frames, wherein each fingerprint of the plurality of second fingerprints is generated based on a respective video frame of the plurality of second video frames, each fingerprint comprising a second coarse-grain fingerprint element and a second fine-grain fingerprint element, the second coarse-grain fingerprint element being generated from a reduced-sized version of the respective video frame that is smaller than the version of the respective video frame used to generate the second fine-grain fingerprint element; comparing the plurality of first fingerprints to the plurality of second fingerprints to generate similarity scores representing how similar each of the plurality of first video frames are to the plurality of second video frames; identifying one or more candidate clip matches for the first video clip based on the similarity scores; selecting a best match for the first video clip in the second video footage from the one or more candidate clip matches for the first video clip; and presenting the best match for the first video clip on a user interface of the data processing system.

2. The data processing system of claim 1, wherein to compare the plurality of first fingerprints to the plurality of second fingerprints to generate the similarity scores, the machine-readable medium includes further executable instructions that, when executed by the processor, cause the data processing system to perform operations comprising: generating a score matrix comparing each respective fingerprint of the plurality of first fingerprints with each respective fingerprint of the plurality of second fingerprints, each cell of the score matrix comprising a similarity score representing a similarity of a respective fingerprint of the plurality of first fingerprints and a respective fingerprint of the plurality of second fingerprints.

3. The data processing system of claim 2, wherein the score matrix comprises a sparse matrix, and wherein the machine-readable medium includes further executable instructions that, when executed by the processor, cause the data processing system to perform operations comprising: applying a filter to the respective fingerprint of the plurality of first fingerprints and the respective fingerprint of the plurality of second fingerprints associated with each cell of the score matrix to determine

whether a similarity threshold is exceeded; determining a respective similarity score for each cell of the score matrix for which the similarity threshold is exceeded; and storing the respective similarity score in each cell for which the similarity score was determined.

4. The data processing system of claim 2, wherein the machine-readable medium includes further executable instructions that, when executed by the processor, cause the data processing system to perform operations comprising: generating a pseudo-probability matrix based on the score matrix, the pseudo-probability matrix including a probability that the respective fingerprint of the plurality of first fingerprints associated with a respective cell of the pseudo-probability matrix is a match for the respective fingerprint of the plurality of second fingerprints.

5. The data processing system of claim 4, wherein the machine-readable medium includes further executable instructions that, when executed by the processor, cause the data processing system to perform operations comprising: for each respective frame of the plurality of first video frames, selecting a respective frame of the plurality of second video frames having a highest probability that the respective frame of the plurality of second video frames is a match for the respective frame of the plurality of first video frames.

6. The data processing system of claim 5, wherein the machine-readable medium includes further executable instructions that, when executed by the processor, cause the data processing system to perform operations comprising: identifying the one or more candidate clip matches by analyzing the similarity scores of cells along a diagonal of the score matrix in which the respective frame having the highest probability of matching each respective frame of the plurality of first video frames resides.

7. The data processing system of claim 6, wherein, to select the best match for the first video clip in the second video footage from the one or more candidate clip matches for the first video clip, the machine-readable medium includes further executable instructions that, when executed by the processor, cause the data processing system to perform operations comprising: calculating a matching score for each clip of the one or more candidate clip matches based on the similarity scores of video frames comprising each clip of the one or more candidate clip matches; and selecting the best match from the one or more candidate clip matches, the best match having a matching score indicative of the best match being a better match for the first video clip than other candidate clips of the one or more candidate clip matches.

8. The data processing system of claim 4, wherein to compare the plurality of first fingerprints to the plurality of second fingerprints to generate similarity scores the machine-readable medium includes further executable instructions that, when executed by the processor, cause the data processing system to perform operations comprising: generating a normalized pseudo-probability matrix based on the pseudo-probability matrix by modifying values of columns of the pseudo-probability matrix so that a total of the values of each column equals one hundred percent.

9. The data processing system of claim 8, wherein the machine-readable medium includes further executable instructions that, when executed by the processor, cause the data processing system to perform operations comprising: averaging values of each cell of the normalized pseudo-probability matrix with a corresponding cell of the pseudo-probability matrix to determine an overall probability score for each cell of the pseudo-probability matrix.

10. The data processing system of claim 8, wherein the machine-readable medium includes further executable instructions that, when executed by the processor, cause the data processing system to perform operations comprising: determining a weighted average values of each cell of the normalized pseudo-probability matrix with a corresponding cell of the pseudo-probability matrix to determine an overall probability score for each cell of the pseudo-probability matrix.

11. A method in a data processing system for matching video content, the method comprising: obtaining first video footage that includes a plurality of first video frames of a first video clip; obtaining second video footage that includes a plurality of second video frames; analyzing the first video footage to generate a plurality of first fingerprints representing each frame of the plurality of

first video frames, wherein each fingerprint of the plurality of first fingerprints is generated based on a respective video frame of the plurality of first video frames, each fingerprint comprising a first coarse-grain fingerprint element and a first fine-grain fingerprint element, the first coarse-grain fingerprint element being generated from a reduced-sized version of the respective video frame that is smaller than a version of the respective video frame used to generate the first fine-grain fingerprint element, wherein generating the plurality of first fingerprints includes: generating the first coarse-grain fingerprint element based on a frame of the first video footage by: resizing the frame of the first video footage to a first standard image size to generate a first resized frame; subdividing the first resized frame into a plurality of cells; and analyzing each of the plurality of cells using a folding locality-sensitive hashing (LSH) algorithm to generate a plurality of hash values comprising a hash value for each cell of the plurality of cells; and generating the first fine-grain fingerprint element based on a frame of the first video footage by: generating a first vector by reducing a dimensionality of a first vector representation of the frame; generating a second vector orthogonal to a vector of ones by subtracting a component of the first vector in a direction of the vector of ones from the component of the first vector; and converting the second vector into a bit string representing values of elements of the second vector; analyzing the second video footage to generate a plurality of second fingerprints representing each frame of the plurality of second video frames, wherein each fingerprint of the plurality of second fingerprints is generated based on a respective video frame of the plurality of second video frames, each fingerprint comprising a second coarse-grain fingerprint element and a second fine-grain fingerprint element, the second coarse-grain fingerprint element being generated from a reduced-sized version of the respective video frame that is smaller than the version of the respective video frame used to generate the second fine-grain fingerprint element; comparing the plurality of first fingerprints to the plurality of second fingerprints to generate similarity scores representing how similar each of the plurality of first video frames are to the plurality of second video frames; identifying one or more candidate clip matches for the first video clip based on the similarity scores; selecting a best match for the first video clip in the second video footage from the one or more candidate clip matches for the first video clip; and presenting the best match for the first video clip on a user interface of the data processing system.

12. The method of claim 11, wherein comparing the plurality of first fingerprints to the plurality of second fingerprints to generate the similarity scores further comprises: generating a score matrix comparing each respective fingerprint of the plurality of first fingerprints with each respective fingerprint of the plurality of second fingerprints, each cell of the score matrix comprising a similarity score representing a similarity of a respective fingerprint of the plurality of first fingerprints and a respective fingerprint of the plurality of second fingerprints; applying a filter to the respective fingerprint of the plurality of first fingerprints and the respective fingerprint of the plurality of second fingerprints associated with each cell of the score matrix to determine whether a similarity threshold is exceeded; determining a respective similarity score for each cell of the score matrix for which the similarity threshold is exceeded; and storing the respective similarity score in each cell for which the similarity score was determined.

13. The method of claim 12, further comprising: for each respective frame of the plurality of first video frames, selecting a respective frame of the plurality of second video frames having a highest probability that the respective frame of the plurality of second video frames is a match for the respective frame of the plurality of first video frames.

14. The method of claim 13, further comprising: identifying the one or more candidate clip matches by analyzing the similarity scores of cells along a diagonal of the score matrix in which the respective frame having the highest probability of matching each respective frame of the plurality of first video frames resides.

15. The method of claim 14, wherein selecting the best match for the first video clip in the second video footage from the one or more candidate clips matches for the first video clip further

comprises: calculating a matching score for each clip of the one or more candidate clip matches based on the similarity scores of video frames comprising each clip of the one or more candidate clip matches; and selecting the best match from the one or more candidate clip matches, the best match having a matching score indicative of the best match being a better match for the first video clip than other candidate clips of the one or more candidate clip matches.

16. A machine-readable medium on which are stored instructions that, when executed, cause a processor of a programmable device to perform operations of: obtaining first video footage that includes a plurality of first video frames of a first video clip; obtaining second video footage that includes a plurality of second video frames; analyzing the first video footage to generate a plurality of first fingerprints representing each frame of the plurality of first video frames, wherein each fingerprint of the plurality of first fingerprints is generated based on a respective video frame of the plurality of first video frames, each fingerprint comprising a first coarse-grain fingerprint element and a first fine-grain fingerprint element, the first coarse-grain fingerprint element being generated from a reduced-sized version of the respective video frame that is smaller than a version of the respective video frame used to generate the first fine-grain fingerprint element, wherein generating the plurality of first fingerprints includes: generating the first coarse-grain fingerprint element based on a frame of the first video footage by: resizing the frame of the first video footage to a first standard image size to generate a first resized frame; subdividing the first resized frame into a plurality of cells; and analyzing each of the plurality of cells using a folding locality-sensitive hashing (LSH) algorithm to generate a plurality of hash values comprising a hash value for each cell of the plurality of cells; and generating the first fine-grain fingerprint element based on a frame of the first video footage by: generating a first vector by reducing a dimensionality of a first vector representation of the frame; generating a second vector orthogonal to a vector of ones by subtracting a component of the first vector in a direction of the vector of ones from the component of the first vector; and converting the second vector into a bit string representing values of elements of the second vector; analyzing the second video footage to generate a plurality of second fingerprints representing each frame of the plurality of second video frames, wherein each fingerprint of the plurality of second fingerprints is generated based on a respective video frame of the plurality of second video frames, each fingerprint comprising a second coarse-grain fingerprint element and a second fine-grain fingerprint element, the second coarse-grain fingerprint element being generated from a reduced-sized version of the respective video frame that is smaller than the version of the respective video frame used to generate the second fine-grain fingerprint element; comparing the plurality of first fingerprints to the plurality of second fingerprints to generate similarity scores representing how similar each of the plurality of first video frames are to the plurality of second video frames; identifying one or more candidate clip matches for the first video clip based on the similarity scores; selecting a best match for the first video clip in the second video footage from the one or more candidate clip matches for the first video clip; and presenting the best match for the first video clip on a user interface of the programmable device.

17. The machine-readable medium of claim 16, wherein, to compare the plurality of first fingerprints to the plurality of second fingerprints generate the similarity scores, the machine-readable medium includes further executable instructions that, when executed by the processor, cause the programmable device to perform operations comprising: generating a score matrix comparing each respective fingerprint of the plurality of first fingerprints with each respective fingerprint of the plurality of second fingerprints, each cell of the score matrix comprising a similarity score representing a similarity of a respective fingerprint of the plurality of first fingerprints and a respective fingerprint of the plurality of second fingerprints; applying a filter to the respective fingerprint of the plurality of first fingerprints and the respective fingerprint of the plurality of second fingerprints associated with each cell of the score matrix to determine whether a similarity threshold is exceeded; determining a respective similarity score for each cell of the score matrix for which the similarity threshold is exceeded; and storing the respective similarity score in

each cell for which the similarity score was determined.

18. The machine-readable medium of claim 17, wherein the machine-readable medium includes further executable instructions that, when executed by the processor, cause the programmable device to perform operations comprising: for each respective frame of the plurality of first video frames, selecting a respective frame of the plurality of second video frames having a highest probability that the respective frame of the plurality of second video frames is a match for the respective frame of the plurality of first video frames; identifying the one or more candidate clip matches by analyzing the similarity scores of cells along a diagonal of the score matrix in which the respective frame having the highest probability of matching each respective frame of the plurality of first video frames resides; calculating a matching score for each clip of the one or more candidate clip matches based on the similarity scores of video frames comprising each clip of the one or more candidate clip matches; and selecting the best match from the one or more candidate clip matches, the best match having a matching score indicative of the best match being a better match for the first video clip than other candidate clips of the one or more candidate clip matches.
